

eRAN

D-MIMO (TDD) Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 eRAN22.1 01 (2026-03-10)

This issue does not include any changes.

1.2 eRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to eRAN21.1 01 (2025-03-10).

Technical Changes

None

Editorial Changes

Added an impact relationship between inter-cell downlink D-MIMO and downlink FDD+TDD CA. For details, see [6.3.2.3 Function Impacts](#).

Added an impact relationship between downlink D-MIMO and precise AMC. For details, see [4.3.2.3 Function Impacts](#).

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding

(MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

Feature Differences Between RATs

The feature difference section only describes differences in switches or principles.

Unless otherwise stated, descriptions in this document apply to all RATs. If a description does not apply to all RATs, the specific RAT that it does apply to will be stated.

For example, in the statement "TDD cells are compatible with enhanced MU-MIMO", "TDD cells" indicates that this function cannot be used in non-TDD cells.

2.2 Applicable RAT

This document applies to TDD.

2.3 Features in This Document

This document describes the following TDD features.

Feature ID	Feature Name	Chapter/Section
TDLEOFD-111505	DL D-MIMO	4 Intra-Cell D-MIMO
TDLEOFD-11150501	Distributed MIMO	4 Intra-Cell D-MIMO
TDLEOFD-11150502	Intra-BBU D-MIMO	4 Intra-Cell D-MIMO
TDLEOFD-11150504	D-MIMO by Macro RRUs	4 Intra-Cell D-MIMO
TDLEOFD-11150505	D-MIMO Within a LampSite Cell	4 Intra-Cell D-MIMO

Feature ID	Feature Name	Chapter/Section
TDLEOFD-121501	Inter-eNodeB DL D-MIMO (Trial)	5 Inter-eNodeB D-MIMO
TDLEOFD-130501	Inter-Cell DL D-MIMO	6 Inter-Cell D-MIMO
TDLEOFD-13050101	Distributed MIMO	6 Inter-Cell D-MIMO
TDLEOFD-13050102	Intra-BBU D-MIMO	6 Inter-Cell D-MIMO
TDLEOFD-13050103	D-MIMO by Macro RRUs	6 Inter-Cell D-MIMO
TDLEOFD-13050104	D-MIMO by pRRUs	6 Inter-Cell D-MIMO

The following table lists the mapping between subfeatures and parent features. The license is deployed on the parent features.

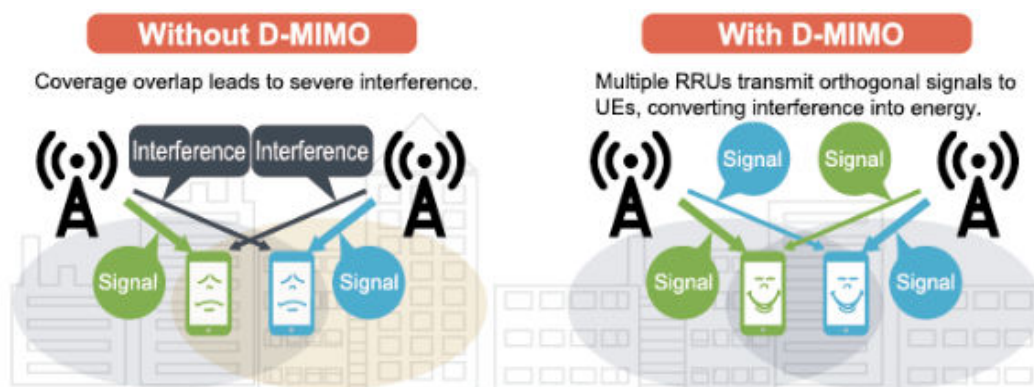
Parent Feature	Subfeature	Scenario
TDLEOFD-111505 DL D-MIMO	TDLEOFD-11150501 Distributed MIMO	Basic functions
TDLEOFD-111505 DL D-MIMO	TDLEOFD-11150502 Intra-BBU D-MIMO	Basic functions
TDLEOFD-111505 DL D-MIMO	TDLEOFD-11150504 D-MIMO by Macro RRUs	Outdoor macro-macro scenarios
TDLEOFD-111505 DL D-MIMO	TDLEOFD-11150505 D-MIMO Within a LampSite Cell	Indoor LampSite scenarios
TDLEOFD-130501 Inter-Cell DL D-MIMO	TDLEOFD-13050101 Distributed MIMO	Basic functions
TDLEOFD-130501 Inter-Cell DL D-MIMO	TDLEOFD-13050102 Intra-BBU D-MIMO	Basic functions
TDLEOFD-130501 Inter-Cell DL D-MIMO	TDLEOFD-13050103 D-MIMO by Macro RRUs	Outdoor macro-macro scenarios
TDLEOFD-130501 Inter-Cell DL D-MIMO	TDLEOFD-13050104 D-MIMO by pRRUs	Indoor LampSite scenarios

3 General Principles

3.1 Overview

Background

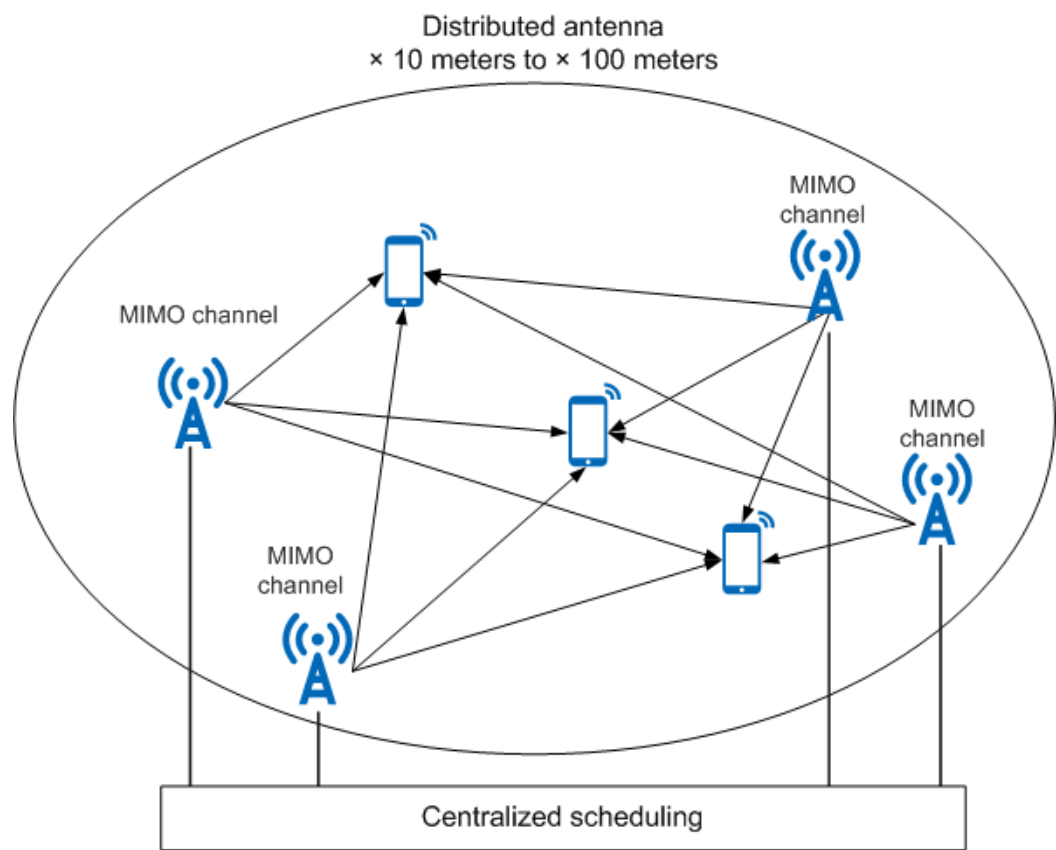
As the number of users and traffic volume increase sharply, the site density rises considerably. Site densification improves the network capacity. However, it also increases the overlapping coverage areas of sites, and causes user experience to deteriorate at the cell edge. Distributed MIMO (D-MIMO) is introduced to mitigate interference from neighboring cells in overlapping coverage areas and improve the experience of cell edge UEs and the downlink user experience of the whole cell.



Introduction

D-MIMO is based on a centralized scheduling+distributed RRU architecture illustrated in [Figure 3-1](#). It uses multiple physical antennas to transmit data in a coordinated way. The dispersed spatial locations of antennas significantly improve the spatial channel resolution. This feature offers high antenna array gains and interference suppression gains while improving the downlink user experience and CEU experience.

Figure 3-1 D-MIMO



D-MIMO coordinates scheduling on a per D-MIMO cluster basis. A D-MIMO cluster is a set of physical cells (generated by RRUs) that compose one or more cells. D-MIMO requires that D-MIMO cluster data be planned, including the RRUs in a D-MIMO cluster and the cluster ID. For details about the planning method, contact Huawei engineers.

Category

D-MIMO has three networking modes.

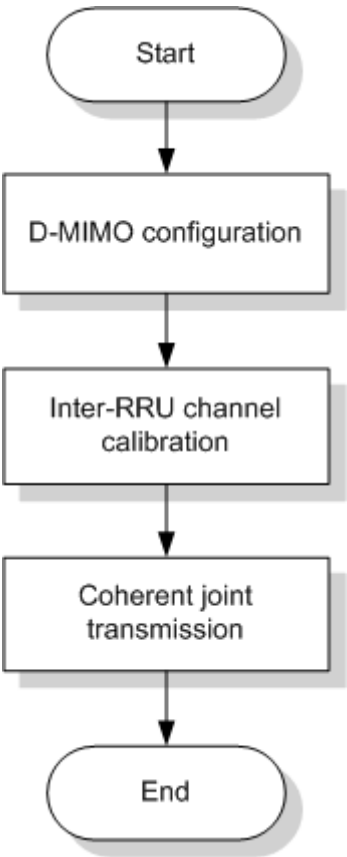
Feature ID	Feature Name	Networking	Introduced In...	Chapter/Section
TDLEOFD-111505	DL D-MIMO	Intra-cell D-MIMO: Multiple RRUs connected to the same BBU and their corresponding physical cells form a logical cell, and a D-MIMO cluster is set up for the logical cell. Cells in the same D-MIMO cluster must have the same physical cell identifier (PCI).	eRAN TDD 11.1	4 Intra-Cell D-MIMO

Feature ID	Feature Name	Networking	Introduced In...	Chapter/Section
TDLEOFD-121501	Inter-eNodeB DL D-MIMO (Trial)	Intra-cell D-MIMO: RRUs connected to different BBUs on the same Cloud BB network and the corresponding cells form a D-MIMO cluster.	eRAN TDD 12.1	5 Inter-eNodeB D-MIMO
TDLEOFD-130501	Inter-Cell DL D-MIMO	Inter-cell D-MIMO: Multiple RRUs connected to the same BBU and their corresponding cells form a D-MIMO cluster. Cells in the same D-MIMO cluster have different PCIs.	eRAN TDD 13.1	6 Inter-Cell D-MIMO

Working Process

Figure 3-2 shows how D-MIMO works.

Figure 3-2 D-MIMO process



1. D-MIMO configuration: involves D-MIMO clusters, feature switches, and related parameters.
2. Inter-RRU channel calibration: D-MIMO adopts the distributed RRU architecture, in which the RRUs in different locations transmit signals with different phases. After inter-RRU channel calibration, the signals sent by multiple RRU channels in the D-MIMO cluster can be combined in the same phase on the UE side.
3. Coherent joint transmission: According to a specific algorithm, joint transmission (JT) UEs in a D-MIMO cluster, which can be single-user joint transmission (SU-JT) or multi-user joint transmission (MU-JT) UEs, are identified to perform coherent joint transmission, improving user experience in the downlink and at the cell edge.

After a D-MIMO cluster is configured and D-MIMO is enabled, the eNodeB periodically initiates intra-cluster inter-RRU channel calibration and coherent joint transmission. D-MIMO is controlled by the following options:

- Intra-cell D-MIMO: **DmimoJTSwitch** option of the **CellAlgoSwitch.DMIMOAlgoSwitch** parameter
- Inter-cell D-MIMO: **InterCellDmimoJTSwitch** option of the **CellAlgoSwitch.DMIMOAlgoSwitch** parameter

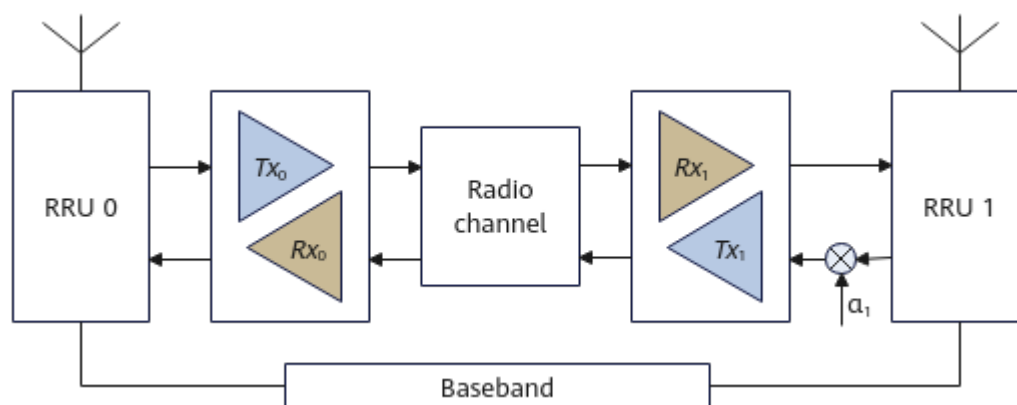
Inter-RRU channel calibration and coherent joint transmission are key technologies of D-MIMO, which will be detailed in the following sections.

3.2 Inter-RRU Channel Calibration

In the distributed RRU architecture used by D-MIMO, the RRUs in different locations transmit signals with different phases. Inter-RRU channel calibration ensures that the RF channels of a D-MIMO cluster have the same delay so that the signals transmitted over different RF channels of the RRUs can be combined in the same phase on the UE side. With inter-RRU channel calibration, the eNodeB adjusts the transmit/receive response ratio of each RF channel of the RRUs to be consistent with that of the reference RF channel.

Figure 3-3 shows the principles of inter-RRU channel calibration.

Figure 3-3 Principles of inter-RRU channel calibration



During inter-RRU channel calibration, the calibration coefficient a_1 is calculated so

that the equation $\frac{Tx_0}{Rx_0} = \frac{a_1 * Tx_1}{Rx_1}$ is satisfied. In the equation, $\frac{Tx_0}{Rx_0}$ indicates the transmit/receive response ratio of the reference RF channel.

The inter-RRU channel calibration procedure is as follows:

1. The intra-RRU channel calibration algorithm is used to perform channel calibration for each RRU.
2. The inter-RRU channel calibration algorithm is used for channel calibration. During inter-RRU channel calibration, the RRUs transmit calibration sequences to each other over the air interface and then calculate the calibration coefficient.
3. Channel compensation is performed based on the calculation results so that the transmit/receive response ratios of all RF channels of RRUs in the D-MIMO cluster are consistent.

NOTE

2T2R RRUs and pRRUs do not support intra-RRU channel calibration. Therefore, only steps 2 and 3 are required for 2T2R RRUs and pRRUs.

In inter-RRU channel calibration, the phases of the signals transmitted over different RRU RF channels are synchronized on the UE side. This increases the strength of the downlink signals received by the UE. For details about how to check whether inter-RRU channel calibration is successful, see [4.4.2 Activation Observation](#).

If a channel calibration fault occurs in a cell in the D-MIMO cluster, the following situations may occur:

- Intra-cell D-MIMO: D-MIMO cells automatically revert to SFN configurations.
- Inter-cell D-MIMO: D-MIMO cells do not revert. In this case, UEs served by the D-MIMO cluster can obtain power superimposition gains but cannot obtain array gains.

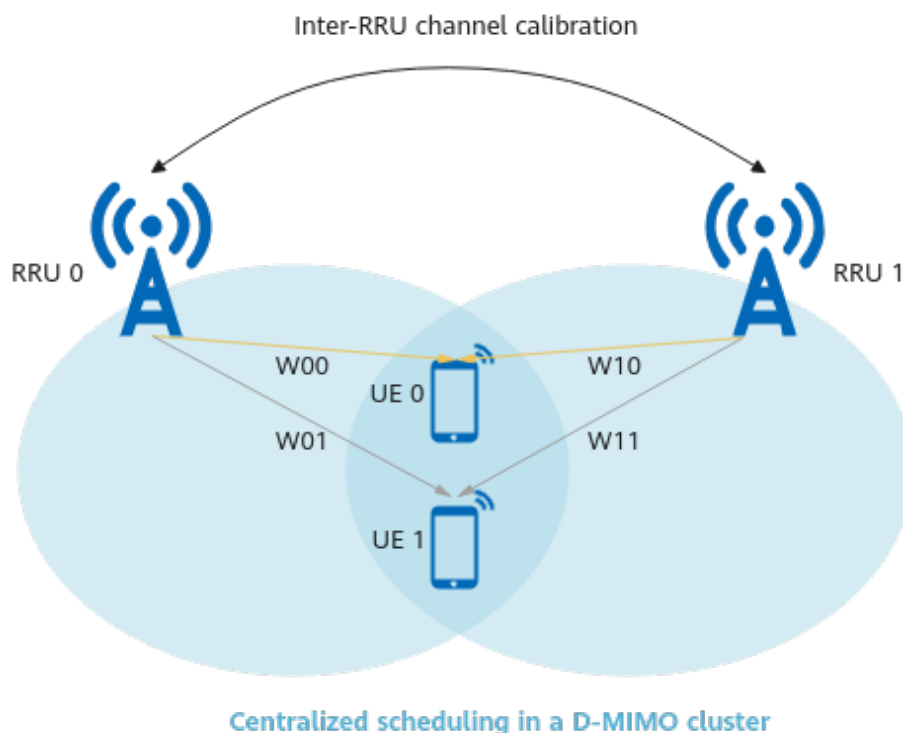
Related Optimization

In eRAN TDD 11.1 or earlier, intra-RRU channel calibration is initiated every 30 minutes. In eRAN TDD 12.0, intra-RRU channel calibration is initiated every 10s if the **QUICK_CHN_CAL_SWITCH** option of the newly added **CellAlgoSwitch.EnhChnCalSwitch** parameter is selected; and intra-RRU channel calibration is still initiated every 30 minutes if this option is deselected. It is recommended that this option be selected in D-MIMO scenarios. 2T2R RRUs and pRRUs do not support intra-RRU channel calibration. Therefore, the setting of the **CellAlgoSwitch.EnhChnCalSwitch** parameter does not take effect on these modules.

3.3 Coherent Joint Transmission

With coherent joint transmission, multiple RRUs jointly transmit signals to the same UE, and these signals are superimposed in phase on the UE side. This process increases the strength of downlink signals received by the UE.

Figure 3-4 Coherent joint transmission



Coherent joint transmission works as follows:

1. Identifying JT UEs: The sounding reference signal (SRS) is used to obtain the uplink RSRP values of signals arriving at each RRU, and calculates the downlink equivalent RSRP values in intra-cell D-MIMO scenarios. The eNodeB periodically identifies the working RRU list for a UE. If the working RRU list for a given UE contains two or more RRUs, the UE is a JT UE.
In inter-cell D-MIMO scenarios, the SRS and event A3 measurement results are used to calculate an equivalent SRS RSRP and event A3 RSRP to select JT UEs. If both the SRS RSRP and event A3 RSRP are available, the eNodeB preferentially selects JT UEs based on the event A3 RSRP.
2. Calculating the beamforming weighting values: The uplink channel coefficients of uplink signals arriving at each RRU are obtained based on the SRS to calculate the joint beamforming weighting values for JT UEs.
Inaccurate beamforming weight estimation will affect beamforming UE pairing performance. The beamforming weights can be estimated more accurately by increasing the SRS SINR of beamforming UEs. When D-MIMO is enabled, it is recommended that target SRS SINR optimization for beamforming UEs be enabled to increase the SRS SINR of beamforming UEs. This function is controlled by the **CellPcAlgo.DMSrsPcSinrOffset** parameter.
 - When the **CellPcAlgo.DMSrsPcSinrOffset** parameter is set to a value greater than 0 and the conditions for triggering this function are met, the eNodeB sets the target SRS SINR for beamforming UEs to the sum of the **CellPcAlgo.SrsPcSinrTarget** parameter value and the **CellPcAlgo.DMSrsPcSinrOffset** parameter value.
 - When the **CellPcAlgo.DMSrsPcSinrOffset** parameter is set to 0 or the conditions for triggering this function are not met, the target SRS SINR

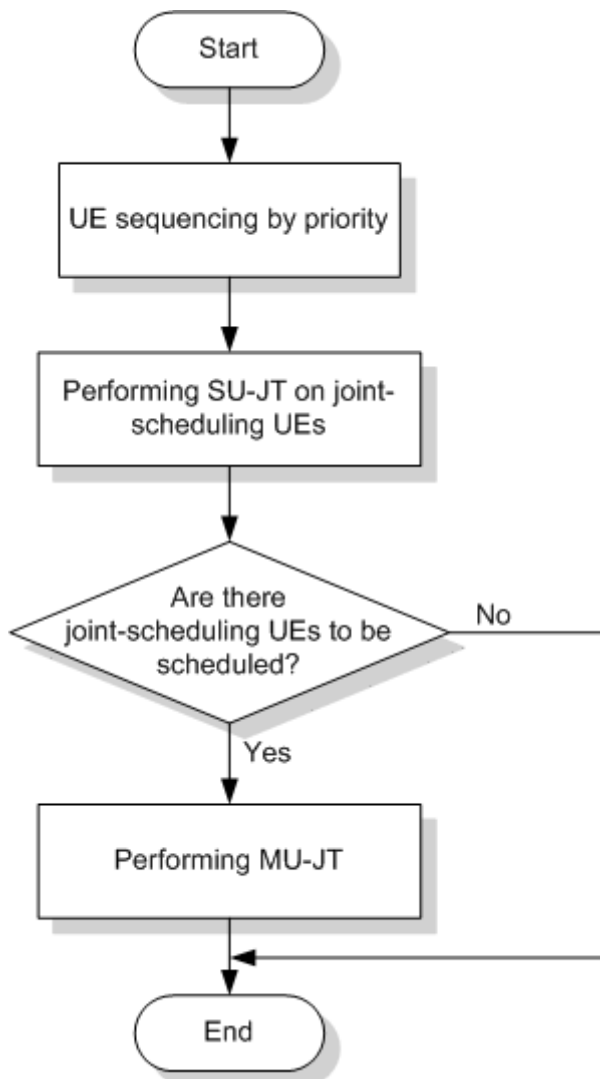
for beamforming UEs is set to the value of the **CellPcAlgo.SrsPcSnrTarget** parameter.

 NOTE

Target SRS SINR optimization for beamforming UEs may slightly increase the number of RRC Connection Reconfiguration messages and the service drop rate.

3. Performing SU-JT or MU-JT in each transmission time interval (TTI)

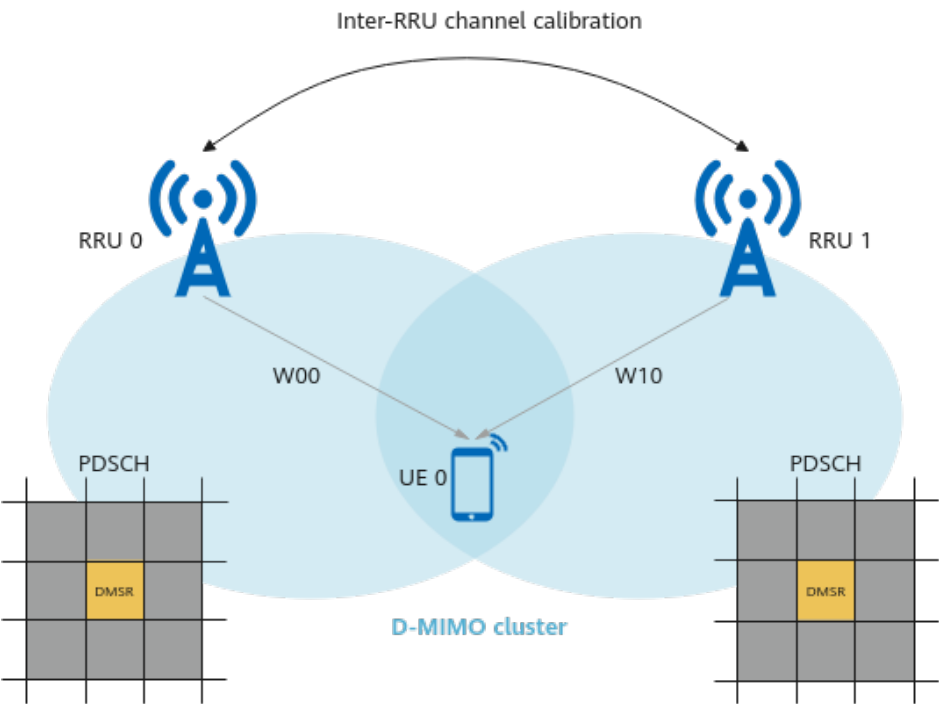
Figure 3-5 JT process



a. SU-JT

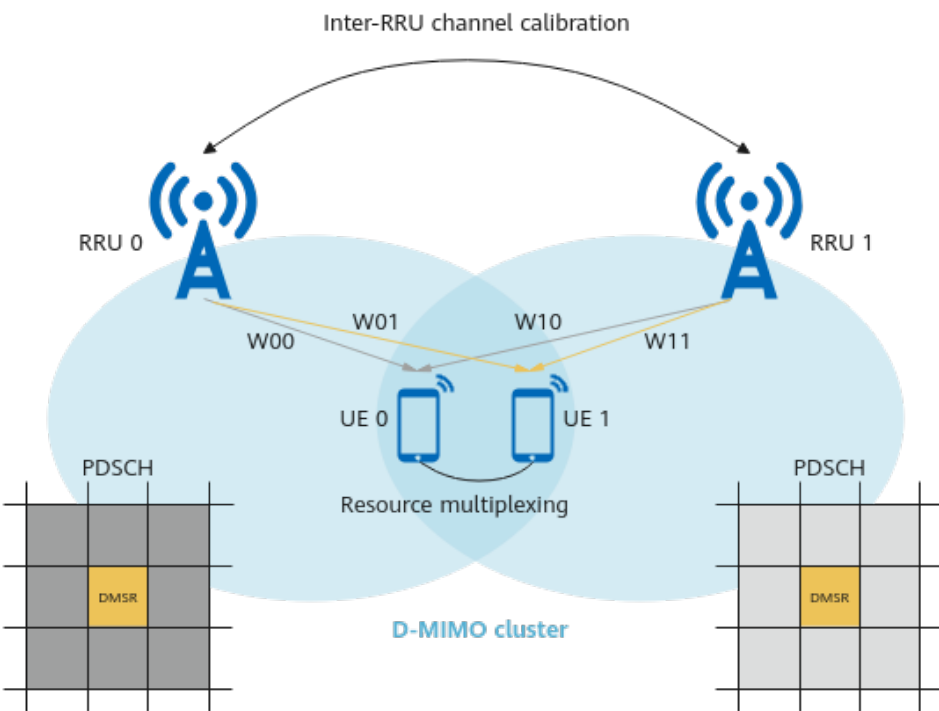
In this scenario, multiple RRUs perform coherent joint transmission to the same UE over multiple antennas, improving the receive signal quality and strength of the UE.

Figure 3-6 SU-JT



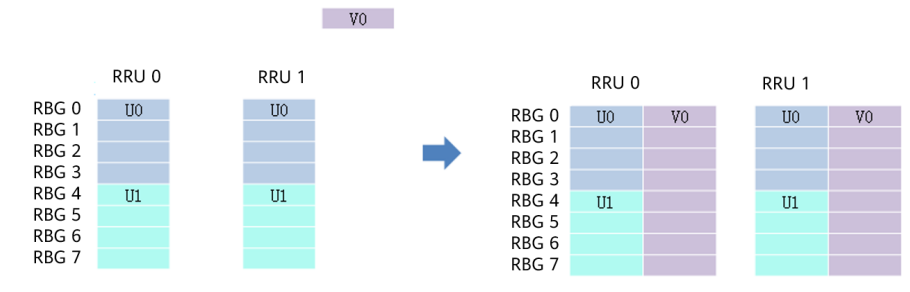
- b. MU-JT
- In this scenario, multiple RRUs perform coherent joint transmission to multiple UEs over multiple antennas, improving the resource reuse efficiency. The weights used for UEs are orthogonal to each other, which mitigates interference between them.

Figure 3-7 MU-JT



Set the **DMIMOcluster.BfWeightNormalizeMode** parameter based on site conditions to achieve optimal MU-JT performance. For details about the setting notes of this parameter, see [4.4.1.1 Data Preparation](#).

Figure 3-8 MU-JT example



The following uses [Figure 3-8](#) as an example to describe the MU-JT process. In the preceding figure, U0 and U1 are the UEs that have been scheduled before MU-JT pairing, and V0 is the UE to be paired. Before MU-JT pairing, only U0 and U1 can be scheduled using RBGs 0 to 7 in a subframe. After MU-JT pairing, V0 and U0 are paired in RBGs 0 to 3, and V0 and U1 are paired in RBGs 4 to 7. In this way, three UEs, including U0, U1, and V0, can be scheduled on the same frequency resources, achieving resource multiplexing and improving spectral efficiency.

When cell-level transmission mode adaptation (TMA) is enabled in non-inter-cell D-MIMO scenarios, more UEs in heavy-load cells enter the beamforming mode for MU pairing, increasing the system capacity. This function is controlled by the **CellBf.WaitPairingLayerThd** parameter.

- When the **CellBf.WaitPairingLayerThd** parameter is set to **0**, this function is not enabled.
- When the **CellBf.WaitPairingLayerThd** parameter is set to a value greater than **0**, the eNodeB performs the following operations: The eNodeB performs cell-level adaptive switching between transmission modes so that more UEs enter the beamforming mode for MU pairing and the system capacity increases if both of the following conditions are met:
 - In heavy load scenarios, the downlink PRB usage of a cell is greater than an internally specified threshold.
 - In multi-layer pairing scenarios, the number of beamforming UEs to be scheduled in the downlink per TTI is greater than or equal to the value of the **CellBf.WaitPairingLayerThd** parameter multiplied by 0.1.

Advanced Multi-User Coordination further increases the MU-JT pairing rate and downlink capacity in intra-cell D-MIMO scenarios when the cell is heavily loaded. This feature is controlled by the **MULTI_UE_COORDINATION_OPT_SW** option of the **CellAlgoSwitch.CoordinationAlgoSwitch** parameter. For details about Advanced Multi-User Coordination, see *Beamforming (TDD)*.

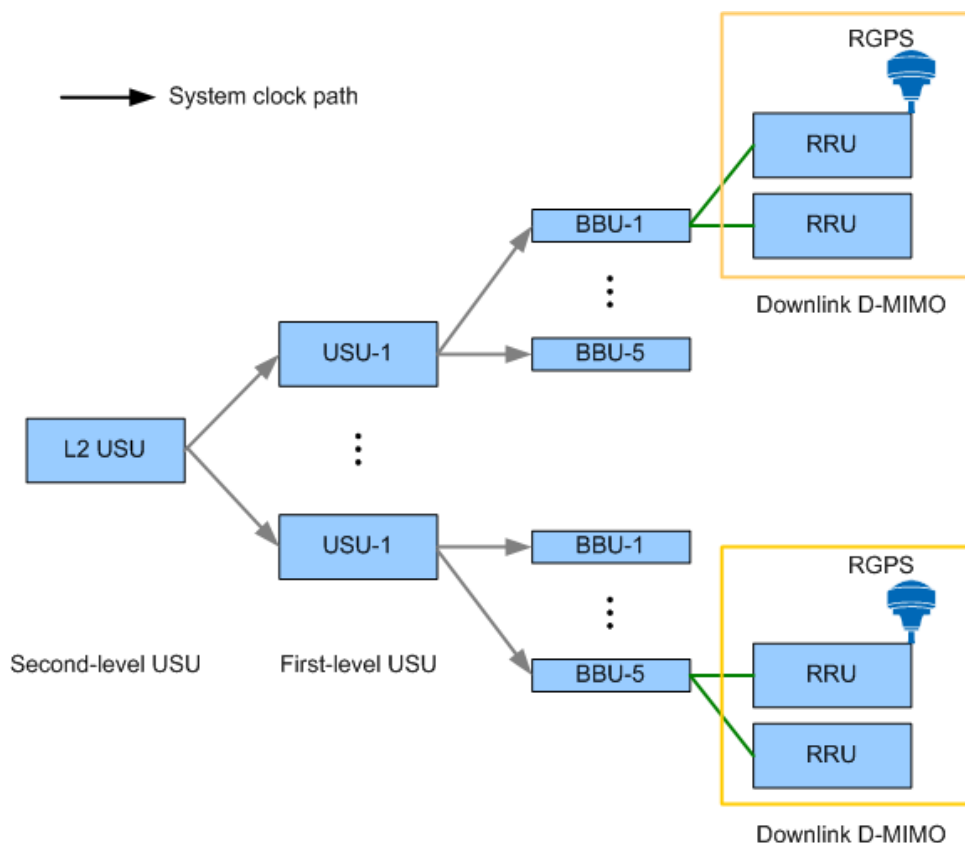
4 Intra-Cell D-MIMO

4.1 Principles

Intra-cell D-MIMO is developed based on SFN. Multiple RRUs connected to a single BBU with overlapping coverage overlaps and their corresponding physical cells constitute a logical cell, and a D-MIMO cluster is set up for the logical cell. The D-MIMO cluster is equivalent to an SFN cell, and therefore the number of RRUs within a D-MIMO cluster equals the number of RRUs serving an SFN cell. Cells in a D-MIMO cluster have the same PCI.

Intra-cell D-MIMO allows RRUs to connect to the same BBU, as shown in [Figure 4-1](#).

Figure 4-1 Networking diagram of intra-cell D-MIMO



Intra-cell D-MIMO requires the following SFN scenarios:

- Outdoor macro-macro SFN scenarios
- Indoor LampSite SFN scenarios

Intra-cell D-MIMO can be used in the following areas:

- Outdoor hotspot areas, such as densely populated urban areas, central business districts, and campus
- Indoor densely populated areas, such as stadiums, airports, railway stations, dining halls, and auditoriums

NOTE

In this document, "macro-macro networking" refers to a combination of macro RRUs. For details, see [RF Modules](#).

4.2 Network Analysis

4.2.1 Benefits

Most Beneficial Scenarios

Intra-cell D-MIMO is recommended when all of the following conditions are met:

- The inter-site distance does not exceed 150 m in outdoor scenarios or 30 m in LampSite scenarios, and there is significant coverage overlap between cells.
- Interference between neighboring cells is severe in LampSite scenarios. This feature is typically suitable in large open areas where intra-frequency neighboring cells are deployed, such as transport hubs and exhibition centers.
- There is line-of-sight (LOS) propagation between RRUs.
- The UE speed is lower than 5 km/h.

Benefits

- In outdoor macro-macro SFN scenarios
D-MIMO increases the average downlink user-experienced rate and the downlink CEU-experienced rate.
- In outdoor micro-micro SFN scenarios
D-MIMO increases the average downlink user-experienced rate and the downlink CEU-experienced rate.
- In indoor LampSite SFN scenarios
D-MIMO increases the average downlink UE-experienced rate.

The combination of intra-cell D-MIMO and advanced multi-user coordination enhances the spatial multiplexing gains. It increases the cell downlink traffic volume and spectral efficiency when the network load is heavy.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- D-MIMO slightly decreases the random access response (RAR) success rate but increases the UE access delay.
- With D-MIMO, when adaptive transmission mode switching is enabled in SFN cells, more UEs switch from TM3 to TM8. As a result, a greater proportion of UEs in the cell use beamforming, including dual-stream beamforming. In addition, dual-stream TM3 UEs report separate CQIs for each of the two data streams. However, dual-stream TM8 UEs report CQIs for only one data stream. Therefore, D-MIMO increases the average CQI value reported by UEs in cells.
- UEs working in dual-stream beamforming must switch to single-stream beamforming before being paired. In this case, the counter related to the number of PRBs used for rank 1 transmission in beamforming is measured. When the proportion of dual-stream beamforming UEs participating in pairing is high, the proportion of the counter related to the number of PRBs used for rank 2 in beamforming decreases.
- D-MIMO enables multiple beamforming UEs in SFN cells to be paired for MU-JT. This enables more UEs to be scheduled within each TTI when many UEs are to be scheduled. However, successful UE scheduling requires that there be sufficient control channel elements (CCEs) available.
 - If CCE resources are sufficient, the total number of CCEs used by each TTI in the downlink increases. This decreases the number of available uplink CCEs, increases the probability of uplink CCE allocation failures, and

affects performance such as the number of uplink scheduled UEs and MCS indexes.

- If CCE resources are insufficient, there will be more CCEs failing to be allocated in each TTI.
- After D-MIMO is enabled, joint-scheduling UEs receive D-MIMO gains and the overall D-MIMO pairing rate improves. The BLER may slightly fluctuate due to the changes in MCS indexes.
- In SFN or D-MIMO scenarios, CRSs are jointly transmitted and PDSCH data is independently transmitted for independent-scheduling UEs. As a result, there may be a mismatch between CRS measurement-based RI values and the RI values supported by PDSCH data. For an independent-scheduling UE working in TM3 rank 2 with a small isolation, RI mismatches lead to consecutive block errors or even cause the MCS index to drop to 0. To solve this problem, the following mechanisms are adopted:
 - When the target RRU works in 2T or 4T mode:

The eNodeB identifies independent-scheduling UEs working in TM3 rank 2 subjected to consecutive block errors caused by RI mismatches and always uses rank 1 to schedule the UEs to ensure UE performance. This function is controlled by the **SfnDLRblerOptSwitch** option of the **CellAlgoSwitch.SfnAlgoSwitch** parameter. In this scenario, one second is required to identify UEs subjected to RI mismatches.
 - When the target RRU works in 8T mode:

UEs working in TM3 rank 2 subjected to consecutive block errors caused by RI mismatches can be adaptively switched to TM8 through TMA to prevent RI mismatches. In this scenario, approximately 2 seconds are required to switch UEs subjected to RI mismatches to TM8.

The MCS index and traffic volume decrease within the occurrence time of an RI mismatch issue if the time is shorter than the preceding time (for example, a UE working in TM3 rank 2 switches to independent scheduling and immediately to joint scheduling).

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
TDLEOFD-111505	DL D-MIMO	LT1SDLMIMO00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Single-stream beamforming	BfSwitch option of the CellAlgoSwitch.BfAlgoSwitch parameter	<i>Beamforming (TDD)</i>	• In outdoor macro-macro scenarios, single-stream beamforming is a prerequisite function. • In indoor LampSite scenarios, single-stream beamforming is not a prerequisite function and therefore the corresponding license is not required. However, the BfSwitch option of the CellAlgoSwitch.BfAlgoSwitch parameter must be selected.
MU beamforming	MuBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter	<i>Beamforming (TDD)</i>	• In outdoor macro-macro scenarios, MU beamforming is a prerequisite function. • In indoor LampSite scenarios, MU beamforming is not a prerequisite function and therefore the corresponding license is not required. However, the MuBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter must be selected.
SFN	• CellAlgoSwitch.SfnULSchSwitch • CellAlgoSwitch.SfnDLSchSwitch	<i>SFN</i>	N/A

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Extended CP	• Cell.UICyclicPrefix • Cell.DICyclicPrefix	<i>Extended CP</i>	None
High speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>	None
Out-of-band relay	CellAlgoSwitch.RelaySwitch	<i>Relay</i>	None
Massive MIMO introduction	Service-based dynamic SRS allocation is controlled by the SrvBasedSRSAdjAlgo option of the SRSCfg.SrsCfgPolaritySwitch parameter. Other functions are free from switch control.	<i>Massive MIMO Basics (TDD)</i>	None
Uplink SU-MIMO	CellAlgoSwitch.UlSuMimoAlgoSwitch	<i>MIMO</i>	None
WTTx turbo beamforming	• TurboBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter • AntSelEnhanceBfSwitch option of the CellAlgoSwitch.BfAlgoSwitch parameter	<i>WTTx Turbo Beamforming (TDD)</i>	None
SRS interference avoidance	WTTxSRSIntrfAvoidanceSw option of the SRSCfg.SrsCfgPolaritySwitch parameter	<i>Massive MIMO Optimization in WTTx Scenarios (TDD)</i>	None

Function Name	Function Switch	Reference	Description
Energy saving based on proactive scheduling	SymbolPwrSaving.TrigBndlSchDLAv gPrbThld	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on proactive scheduling cannot be enabled together with intra-cell D-MIMO.
RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None
Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and rank-4 closed-loop TM9	TM3R2_TO_TM9R4_THLD_ADAPT_SW option of the CellBfMimoParaCfg.BfMimoAlgoOptSwitch parameter	<i>Beamforming (TDD)</i>	Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and rank-4 closed-loop TM9 does not take effect if the Cell.MultiRruCellFlag parameter is set to BOOLEAN_TRUE .
Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and dual-stream beamforming	TM3R2_TO_DUALBF_THLD_ADAPT_SW option of the CellBfMimoParaCfg.BfMimoAlgoOptSwitch parameter	<i>Beamforming (TDD)</i>	Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and dual-stream beamforming does not take effect if the Cell.MultiRruCellFlag parameter is set to BOOLEAN_TRUE .
Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>	None

Function Name	Function Switch	Reference	Description
Adaptive MU beamforming	CellBf.AdaptMubfStartThld set to a value other than 255	<i>Beamforming (TDD)</i>	None
MuteUppTsSymbForSrsSw	MuteUpptsSymbForSrsSw option of the SRSCfg.SrsCfgPolicySwitch parameter	None	The switch specified by the MuteUpptsSymbForSrsSw option cannot be turned on together with the switch for intra-cell D-MIMO.
Single-cell intelligent power control energy saving	INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Inter-RRU channel calibration will fail if RF channel intelligent shutdown is enabled.
Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	If a cell in an inter-cell D-MIMO cluster enters the intelligent power-off of carriers in the same coverage state, inter-RRU channel calibration fails and the inter-cell D-MIMO cluster is disabled.
Low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	When a cell within a D-MIMO cluster is in low power consumption mode, it cannot serve as a cooperating cell for joint transmission and inter-RRU channel calibration may fail.

Function Name	Function Switch	Reference	Description
MU beamforming	MuBfSwitch option of the CellAlgoSwitch . MuBfAlgoSwitch parameter	<i>Beamforming (TDD)</i>	When D-MIMO is enabled, multi-user spatial multiplexing is implemented using D-MIMO, and the performance counters and performance monitoring items related to MU beamforming are no longer measured.
DRX	CellDrxPara.DrxAlgSwitch	<i>DRX and Signaling Control</i>	If DRX is enabled by setting the CellDrxPara.DrxAlgSwitch parameter, UEs do not send SRS to the eNodeB during DRX sleep time. As a result, the instantaneous beamforming weighting values of D-MIMO UEs cannot be updated promptly, deteriorating D-MIMO performance.
Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	D-MIMO allows multiple RF modules working in the same frequency band in the same geographical area to serve the same cell, while power sharing between carriers is performed based on physical cells that share one RF module. When both functions are enabled, dynamic power sharing cannot provide maximum gains.
SFN	None	<i>SFN</i>	When D-MIMO is enabled, interference suppression between UEs is implemented using D-MIMO, and the performance counters and performance monitoring items related to inter-RRU coordinated beamforming are no longer measured in SFN scenarios.

Function Name	Function Switch	Reference	Description
Adaptive switching between beamforming and MIMO	CellBfMimoParaCfg.BfMimoAdaptiveSwitch	<i>Beamforming (TDD)</i>	- When the cell load is light, transmission mode adaptation (TMA) oriented for optimal single-link performance is adopted. - To increase the cell capacity when the cell load is heavy, TMA oriented for optimal single-link performance cannot be used. Cell-level TMA is recommended when the cell load is heavy and multiple layers are available for pairing.
Preferential use of resource allocation type 1	CellDlSchAlgo.RbgAllocStrategy set to TYPE1_FIRST	<i>Downlink Scheduling</i>	When D-MIMO is enabled, preferential use of resource allocation type 1 does not take effect.
MCS selection for transmission of small amounts of data	SmallPktMcsSelectAlgoSwitch option of the CellAlgoSwitch.DlSchSwitch parameter	<i>Downlink Scheduling</i>	MCS selection for transmission of small amounts of data does not take effect for D-MIMO UEs in the joint transmission state.
pRRU deep dormancy in intelligent mode	EnodebMpruEs.MpruDormancyDIEarfcn	<i>Energy Conservation and Emission Reduction</i>	Downlink D-MIMO no longer takes effect when some pRRUs enter the dormancy state.
Precise AMC	PreciseAmcSwitch option of the CellAlgoSwitch.E mimoSwitch parameter	<i>Downlink Scheduling</i>	Precise AMC does not take effect for D-MIMO UEs in the joint transmission state.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The following base stations are compatible with this feature:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

NOTE

- The compatible site type differences of its sub-features are as follows:
 - TDLEOFD-11150501 Distributed MIMO is supported by macro and LampSite base stations.
 - TDLEOFD-11150502 Intra-BBU D-MIMO is supported by macro base stations.
 - TDLEOFD-11150504 D-MIMO by Macro RRUs is supported by macro base stations.
 - TDLEOFD-11150505 D-MIMO Within a LampSite Cell is supported by LampSite base stations.
- For details about the available bandwidths and configuration restraints in LampSite scenarios, see *LampSite Solution Description*, *LampSite BBU Technical Specifications*, *LampSite pRRU&RRU Technical Specifications*, and *LampSite RHUB Technical Specifications* in 3900 & 5900 Series Base Station Product Documentation.

Boards

- Baseband processing unit (BBP)

The UBBPe4, UBBPd6, UBBPd9, or UBBPe6 series board is required for setting up cells and connecting to RRUs corresponding to the cells through optical fibers. Cells in a D-MIMO cluster can be set up on different BBPs but must be in the same BBU.

The UBRlb board does not support D-MIMO.

Table 4-1 Cell specifications supported by different types of BBPs

Board	Transmit/ Receive Mode	Maximum Number of 10 MHz or 20 MHz Cells Supported
UBBPd6 or UBBPd9	2T2R or 4T4R	• 6 when the board does not function as a co-processing board • 3 when the board also functions as a co-processing board
	8T8R	• 6 when the board does not function as a co-processing board • 3 when the board also functions as a co-processing board
UBBPd4	2T2R or 4T4R	6. The board cannot function as a co-processing board.
	8T8R	3. The board cannot function as a co-processing board.
UBBPd6	2T2R or 4T4R	12
	8T8R	6

The outdoor BBU3910A does not support D-MIMO or enhanced channel calibration.

- Co-processing board

This feature requires a co-processing board to calculate D-MIMO co-processing resources. The configuration principles of co-processing boards are as follows:

- The UBBPd6, UBBPe6, or UBBPd9 must be used.
- A maximum of 72 antennas and 6 clusters are supported when a UBBPd is used. For example:
 - Macro 8T8R scenarios: 8 x 3 D-MIMO clusters x 3 RRUs per D-MIMO cluster = 72 antennas
 - Macro 4T4R scenarios: 4 x 6 D-MIMO clusters x 3 RRUs per D-MIMO cluster = 72 antennas
 - Micro 2T2R scenarios: 2 x 6 D-MIMO clusters x 6 RRUs per D-MIMO cluster = 72 antennas4T and 8T D-MIMO clusters can use the same co-processing resources.
- A maximum of 144 antennas and 12 clusters are supported when a UBBPe is used. For example:
 - Macro 8T8R scenarios: 8 x 6 D-MIMO clusters x 3 RRUs per D-MIMO cluster = 144 antennas
 - Macro 4T4R scenarios: 4 x 12 D-MIMO clusters x 3 RRUs per D-MIMO cluster = 144 antennas
 - Micro 2T2R scenarios: 2 x 12 D-MIMO clusters x 6 RRUs per D-MIMO cluster = 144 antennas4T and 8T D-MIMO clusters can use the same co-processing resources.
- If a BBU3900 is used, a co-processing board must be configured in slot 2 or 3. If a BBU3910 is used, there are no constraints on slot configuration.
- A co-processing board cannot be configured as a dedicated scheduling board.

 NOTE

A dedicated scheduling board is a BBP that processes only signaling, instead of service data, in dedicated scheduling mode. For details, see *Cloud BB Overview*.

- A UBBPd cannot be configured as a co-processing board when four or more physical cells are bound to the board.
- A UBBPe cannot be configured as a co-processing board when four or more 8T8R 20 MHz physical cells are configured on the board and CPRI compression is not enabled or when seven or more 4T4R or 8T8R physical cells are configured in other scenarios.
- An FDD+TDD BBP cannot be configured as a co-processing board for D-MIMO.
- If a BBP is configured as a co-processing board, it is impossible to set up both 4T4R and 8T8R cells on this BBP.

- A co-processing board uses the cell frame offset specified by the **CellFrameOffset**. **FrameOffset** parameter on the corresponding BBP. To ensure coordinated scheduling in a cluster, the cell frame offset of the cluster needs to be the same as that used by the co-processing board.

RF Modules

Category	RRU Model	Number of Channels	Supported Frequency Band	Scenario
Macro RRU	RRU3252	4T4R	Band 38	Outdoor macro-macro
Macro RRU	RRU3256	4T4R	Band 42 or band 43	Outdoor macro-macro
Macro RRU	RRU3278	8T8R	Band 42 or band 43	Outdoor macro-macro
Macro RRU	RRU3279	8T8R	Band 38	Outdoor macro-macro
pRRU	pRRU3901	2T2R	2.3 GHz	Indoor LampSite
pRRU	pRRU3911	2T2R	2.3 GHz, 2.5 GHz, or 2.6 GHz	Indoor LampSite

NOTE

When the RRU3279 serves two 4T4R sectors by splitting, it does not support D-MIMO.
NR-capable RF modules do not support D-MIMO.

In addition, there are the following constraints:

- The RRUs incapable of D-MIMO do not support enhanced channel calibration.
- RRUs cannot be combined.
- Macro RRUs cannot be cascaded.
- RRUs cannot be configured using the load sharing networking.
- RRUs cannot be configured using the ring networking.
- RRUs cannot be cascaded in macro-macro scenarios.
- The remote RRU deployment distance cannot exceed 20 km.
- No more than three carriers can be configured on the same RRU.
- In FDD+TDD RRU cascading, the upper-level RRU can be a TDD RRU, not an FDD RRU.

Cells

- The cell bandwidth must be 10 MHz or 20 MHz.

- Bands 38, 40, 42, and 43 are supported in outdoor scenarios.
- The 2.3 GHz, 2.5 GHz, and 2.6 GHz bands are supported in indoor LampSite scenarios.

Configuration Restraints in LampSite Scenarios

- D-MIMO cannot be used in pRRU combination scenarios.
- pRRUs belonging to the same D-MIMO cluster must be connected to the same RHUB or connected to two neighboring RHUBs on the same CPRI link.
- No extender is used to extend the Ethernet cable connecting the RHUB to the pRRU.

4.3.4 Networking

In inter-BBU scenarios, D-MIMO is supported only in LTE TDD single-mode or LTE TDD/TD-SCDMA dual-mode networking.

Planning of SFN Networking

- In SFN scenarios, D-MIMO can be deployed directly.
- In non-SFN scenarios, networking must be planned using a D-MIMO cluster planning tool.

Planning of D-MIMO Cluster IDs

D-MIMO cluster IDs need to be planned using a D-MIMO cluster planning tool, which can be obtained from Huawei engineers.

D-MIMO cluster planning must meet the following requirements:

- Basic requirements
 - Cells generated by more than one RRU or more than two pRRUs form a D-MIMO cluster.
 - Each cluster ID must have a unique mod 6 value for each RRU in multi-carrier scenarios.
 - The D-MIMO cluster ID must be unique within a BBU.
 - The IDs of physically neighboring D-MIMO clusters must not have the same mod 24 value.
- Macro-macro scenarios

A D-MIMO cluster allows for two to four RRUs.
- LampSite scenarios
 - A D-MIMO cluster allows for three to six pRRUs.
 - If multiple carriers are configured, both the number of LTE TDD carriers and the LTE TDD bandwidth must be the same for all the pRRUs belonging to the same D-MIMO cluster, regardless of whether D-MIMO is enabled for all the carriers.
 - If more than four pRRUs are configured, the D-MIMO cluster ID must be an even number.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Parameters Used for Activation

For details about data preparation for an SFN cell, see *SFN*. The following tables describe the data preparations for setting up a D-MIMO cluster.

Table 4-2 Parameters in the **CoProcRes** MO

Parameter Name	Parameter ID	Setting Notes
Coordinate Process Resource ID	CoProcRes.CoProcResId	Set this parameter to a value ranging from 0 to 12 .
BaseBand Equipment ID	CoProcRes.BaseBandEqmId	Set this parameter to the ID of the baseband equipment on which the co-processing board is configured. Set this parameter to 255 or to a value ranging from 0 to 23 .
Work Mode	CoProcRes.WorkMode	- This parameter does not need to be configured if only D-MIMO is supported. - Select the COORDINATING_PROCESSING option if D-MIMO and other coordination features are supported.
Bundling Cluster Type	CoProcRes.BundlingCluster-Type	- Set this parameter to DMIMO if only D-MIMO is supported. - Set this parameter only to ADAPTIVE if D-MIMO and other coordination features are supported.

Table 4-3 Parameters in the **DMIMOCluster** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Cluster ID	DMIMOCluster.DMIMOClusterId	Set this parameter to a value ranging from 0 to 65535. • The D-MIMO cluster ID must be an even number if more than four pRRUs are configured in LampSite scenarios. • Each cluster ID must have a unique mod 6 value for each RRU in multi-carrier scenarios. • The D-MIMO cluster ID must be unique within a BBU.
Coordinate Process Resource ID	DMIMOCluster.CoProcResId	Set this parameter to 255 or to a value ranging from 0 to 12. No co-processing resources are specified when this parameter is set to 255.
Beamforming Weight Normalized Mode	DMIMOCluster.BfWeightNormalizeMode	• If this parameter is set to NEBF, paired beamforming UEs achieve optimal performance when residing at the D-MIMO cell edge. • If this parameter is set to PEBF, paired beamforming UEs achieve optimal performance when residing at the center or at a medium distance from the center of the D-MIMO cell. Set this parameter to NEBF in commercial use scenarios.

Table 4-4 Parameters in the **DMIMOClusterCell** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Cluster ID	DMIMOClusterCell.DMIMOClusterId	Set this parameter to a value ranging from 0 to 65535.

Parameter Name	Parameter ID	Setting Notes
Cell ID	DMIMOClusterCell.CellId	Set this parameter to a value ranging from 0 to 255 .
eNodeB ID	DMIMOClusterCell.eNodeBId	Set this parameter to a value ranging from 0 to 1048575 .
Mobile Country Code	DMIMOClusterCell.Mcc	Set this parameter to a value ranging from 000 to 999 .
Mobile Network Code	DMIMOClusterCell.Mnc	Set this parameter to a value ranging from 00 to 99 or from 000 to 999 .

 NOTE

- Only SFN cells and pRRU aggregation cells can be added to a D-MIMO cluster. Only one cell can be added to each D-MIMO cluster.
- The same cell cannot be added to different D-MIMO clusters.
- A given cell can be added to a D-MIMO cluster by the primary operator ID or secondary operator ID but not both. It is recommended that the cell be added to a D-MIMO cluster by the primary operator ID.
- A cell cannot be added to a D-MIMO cluster by both the primary operator ID and the secondary operator ID.

Table 4-5 Parameters in the **CellAlgoSwitch** MO

Parameter Name	Parameter ID	Setting Notes
BF algorithm switch	CellAlgoSwitch.BfAlgoSwitch	Select the BfSwitch option of this parameter.
MUBF Algorithm Switch	CellAlgoSwitch.MuBfAlgoSwitch	Select the MuBfSwitch option of this parameter.
D-MIMO Algorithm Switch	CellAlgoSwitch.DMIMOAlgoSwitch	Inter-RRU channel calibration and joint transmission for D-MIMO are enabled when the DmimoJTSwitch option of this parameter is selected.
Enhanced Channel Calibration Switch	CellAlgoSwitch.EnhChnCalSwitch	It is recommended that the QUICK_CHN_CAL_SWITCH option of this parameter be selected when D-MIMO is enabled.

Parameter Name	Parameter ID	Setting Notes
Coordination Algorithm Switch	CellAlgoSwitch.CoordinationAlgoSwitch	It is recommended that the MULTI_UE_COORDINATION_OPT_SW option of this parameter be selected when D-MIMO is enabled.

Table 4-6 Parameters in the **CellPcAlgo** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Srs Power Control Offset for Sinr Target	CellPcAlgo.DMSrsPcSinrOffset	It is recommended that this parameter be set to 5 when D-MIMO is enabled. Retain the default value 0 when D-MIMO is disabled.

Table 4-7 Parameters in the **CellBf** MO

Parameter Name	Parameter ID	Setting Notes
Wait Pairing Layer Number Threshold	CellBf.WaitPairingLayerThd	- It is recommended that this parameter be set to 30 when the cell load is heavy and the multi-layer pairing rate is high. - It is recommended that this parameter be set to 0 in other situations.

Table 4-8 Parameters in the **CellPdcchAlgo** MO

Parameter Name	Parameter ID	Setting Notes
PDCCH Beamforming Gain Offset	CellPdcchAlgo.PdcchBfGainOffset	It is recommended that this parameter be set to 0 when D-MIMO is enabled and the CellPdcchAlgo.PDCCHAggLvlAdaptStrage parameter is set to STRATEGYBASEDONCOVERAGE . Retain the default value -127 when D-MIMO is disabled.

Table 4-9 Parameters in the **CellBfMimoParaCfg** MO

Parameter Name	Parameter ID	Setting Notes
Transmission Mode Acceleration Switch	CellBfMimoParaCfg.TmAccelerationSwitch	It is recommended that this parameter be set to INITIAL_ACCESS_TO_BF when D-MIMO is enabled. Retain the default value OFF when D-MIMO is disabled.

Table 4-10 Parameters in the **CellDlschAlgo** MO

Parameter Name	Parameter ID	Setting Notes
Downlink High Load SDMA Threshold Offset	CellDlschAlgo.DlHighLoadSdmaThdOffset	It is recommended that this parameter be set to 10 if both D-MIMO and quick entry into beamforming are enabled and to 5 if only D-MIMO is enabled in micro-micro scenarios. Retain the default value 0 when both D-MIMO and quick entry into beamforming are disabled in micro-micro scenarios. This parameter is not required in macro-macro or macro-micro scenarios.

Parameters Used for Optimization

- SRS resource allocation optimization

The SRS resource optimization function is supported in LampSite scenarios to improve D-MIMO performance. This function increases the proportion of UEs allocated with short-period SRS resources, improving beamforming performance for the UEs and enhancing the downlink spectral efficiency of the cell.

The **SRSCfg.SrsResOptSwitch** parameter controls this function and can be configured only when all of the following conditions are met:

- D-MIMO is enabled.
- UBBP boards are manually bound.
- The **SRSCfg.TddSrsCfgMode** parameter is set to **ACCESS_ENHANCED**.
- The **Cell.MultiRruCellMode** parameter is set to **MPRU_AGGREGATION**.

NOTICE

- Modifying the **SRSCfg.SrsResOptSwitch** parameter is a high-risk operation. Therefore, perform this operation with caution.
- The SRS resource optimization function requires the UBBPe/UBBPd6/UBBPd9.
- The UBBPd4 and LBBPd boards do not support SRS resource optimization. If this function is enabled on these boards, physical cells on these boards cannot be activated, and the eNodeB reports ALM-29243 Cell Capability Degraded or ALM-29240 Cell Unavailable. In this situation, the value of the **Work Status** parameter is **The BBP configuration does not support SRS resource optimization** in the **DSP CELL** command output.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following illustrates how to enable D-MIMO for an SFN cell whose local cell ID is 0.

Slot 0	SectorEqmId0		Slot 4
Slot 1	SectorEqmId1		Slot 5
Slot 2	SectorEqmId2		Slot 6
Slot 3	DMIMO Co-Process	UMPT	Slot 7

In this example:

UBBP boards are installed in slots 0 to 3, and the UBBP in slot 3 functions as the co-processing board.

- Four pieces of sector equipment are configured for the cell on a LampSite eNodeB.
- Three pieces of sector equipment are configured for the SFN cell on a macro eNodeB and are bound to UBBP boards in slots 0 to 2, respectively.

For details about how to set up and activate an SFN cell, see *SFN*.

```
//Specifying baseband equipment for the BBP and co-processing board
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=0;
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=1;
ADD BASEBANDEQM: BASEBANDEQMID=2, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=2;
ADD BASEBANDEQM: BASEBANDEQMID=3, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=3;
//(Only for macro eNodeBs) Adding sector equipment and binding baseband equipment for physical cells in
SFN scenarios
ADD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=0, BaseBandEqmId=0;
ADD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=1;
ADD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=2, BaseBandEqmId=2;
//(Only for LampSite eNodeBs) Adding sector equipment groups for LampSite cells and adding sector
equipment to the sector equipment groups
ADD EUSECTOREQMGROUP: LocalCellId=0, SectorEqmGroupId=0;
ADD EUSECTOREQMGROUP: LocalCellId=0, SectorEqmGroupId=1;
ADD EUSECTOREQMGROUP: LocalCellId=0, SectorEqmGroupId=2;
ADD EUSECTOREQMGROUP: LocalCellId=0, SectorEqmGroupId=3;
ADD EUSECTOREQMID2GROUP: LocalCellId=0, SectorEqmGroupId=0, SectorEqmId=0;
ADD EUSECTOREQMID2GROUP: LocalCellId=0, SectorEqmGroupId=1, SectorEqmId=1;
ADD EUSECTOREQMID2GROUP: LocalCellId=0, SectorEqmGroupId=2, SectorEqmId=2;
ADD EUSECTOREQMID2GROUP: LocalCellId=0, SectorEqmGroupId=3, SectorEqmId=3;
//Adding the D-MIMO co-processing resource
ADD COPROCRES: CoProcResId=0, BaseBandEqmId=3, BundlingClusterType=DMIMO;
//Adding a D-MIMO cluster
ADD DMIMOCLUSTER: DMIMOClusterId=0, CoProcResId=0;
//Adding the SFN cell to the D-MIMO cluster
ADD DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=0, eNodeBId=0, Mcc="100", Mnc="01";
//(Optional) Enabling target SRS SINR optimization for beamforming UEs
MOD CELLPCALGO: LocalCellId=0, DMSrsPcSinrOffset=5;
//Turning on the beamforming, MU beamforming, D-MIMO, and enhanced channel calibration switches
MOD CELLALGOSWITCH: LocalCellId=0, BfAlgoSwitch=BfSwitch-1, MuBfAlgoSwitch=MuBfSwitch-1,
DMIMOAlgoSwitch=DmimoJTSwitch-1, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-1;
//(Optional) Turning on the advanced multi-user coordination switch
MOD CELLALGOSWITCH: LocalCellId=0, CoordinationAlgoSwitch= MULTI_UE_COORDINATION_OPT_SW-1;
//(Optional) Setting the PDCCH beamforming gain offset to 0 to enable CCE selection optimization for joint-
scheduling beamforming UEs in scenarios where the coverage-based PDCCH aggregation level adaptation
strategy is used
MOD CELLPDCCHALGO: LocalCellId=0, PdcchBfGainOffset=0;
//(Optional) Setting the threshold for the number of layers to be paired to 30 and enabling cell-level
adaptive switching between transmission modes when the cell load is heavy and the multi-layer pairing
rate is high
MOD CELLBF: LocalCellId=0, WaitPairingLayerThd=30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling target SRS SINR optimization for beamforming UEs
MOD CELLPCALGO: LocalCellId=0, DMSrsPcSinrOffset=0;
//Turning off the beamforming, MU beamforming, D-MIMO, and enhanced channel calibration switches
MOD CELLALGOSWITCH: LocalCellId=0, BfAlgoSwitch=BfSwitch-0, MuBfAlgoSwitch=MuBfSwitch-0,
DMIMOAlgoSwitch=DmimoJTSwitch-0, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-0;
//(Optional) Turning off the advanced multi-user coordination switch
```



```
MOD CELLALGOSWITCH: LocalCellId=0, CoordinationAlgoSwitch= MULTI_UE_COORDINATION_OPT_SW-0;  
//Removing an SFN cell from the D-MIMO cluster  
RMV DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=0, eNodeBId=0, Mcc="100", Mnc="01";  
//Removing the D-MIMO cluster  
RMV DMIMOCLUSTER: DMIMOClusterId=0;  
//(Optional) Restoring the PDCCH beamforming gain offset to disable CCE selection optimization for joint-  
scheduling beamforming UEs in scenarios where the coverage-based PDCCH aggregation level adaptation  
strategy is used  
MOD CELLPDCCHALGO: LocalCellId=0, PdcchBfGainOffset=-127;  
//(Optional) Restoring the threshold for the number of layers to be paired to the value used before D-  
MIMO is enabled when the cell load is heavy and the multi-layer pairing rate is high  
MOD CELLBF: LocalCellId=0, WaitPairingLayerThd=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Observation

- Step 1** Run the **DSP CELL** and **LST CELL** commands to check the cell status of the D-MIMO cluster. The cells in the D-MIMO cluster are working properly if all of the following conditions are met in the command outputs:
- The value of the **Mode of Multi-RRU Cell** parameter is **SFN** or **MPRU_AGGREGATION** in the **LST CELL** command output, indicating that the cell is a multi-sector cell.
 - The value of the **Work Status** parameter is **Normal** in the **DSP CELL** command output for all the RRUs, indicating that the RRUs are working properly.
 - The value of the **Cell instance state** parameter is **Normal** in the **DSP CELL** command output, indicating that the cells have been activated.
- Step 2** Run the **DSP DMIMOCLUSTERCELL** command to check the cell status of the D-MIMO cluster. The D-MIMO cluster is working properly if the value of the **D-MIMO Cluster Cell Status** parameter is **Normal**.
- Step 3** Run the **DSP DMIMOCALIBRATION** command to check the channel calibration status of the D-MIMO cluster. Channel calibration is successful in the D-MIMO cluster if the value of the **D-MIMO Calibration Result** parameter is **Succeeded**.

----End

4.4.3 Network Monitoring

Monitoring Whether D-MIMO Takes Effect

The following counters can be used on the eNodeB side to check whether D-MIMO has taken effect.

Table 4-11 Counters used to observe UE pairing for D-MIMO

Counter Name	Counter Description
L.ChMeas.DMIMO.1Layer.PR	Average number of PRBs that can be paired for D-MIMO
L.ChMeas.DMIMO.2Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 2
L.ChMeas.DMIMO.3Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 3
L.ChMeas.DMIMO.4Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 4
L.ChMeas.DMIMO.5Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 5
L.ChMeas.DMIMO.6Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 6
L.ChMeas.DMIMO.7Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 7
L.ChMeas.DMIMO.8Layer.PairPR	Average number of PRBs paired for D-MIMO at layer 8
Note: • The L.ChMeas.DMIMO.1Layer.PR counter measures the number of PRBs that can be paired for D-MIMO. D-MIMO takes effect if the value of this counter is greater than 0. • The other counters measure the number of PRBs paired for D-MIMO at the corresponding layers.	

Table 4-12 Counters used to observe the proportion and scope of UEs on which D-MIMO takes effect

Counter Name	Counter Description
L.ChMeas.DMIMO.JT.User.Avg	Number of D-MIMO JT UEs in the downlink
L.ChMeas.DMIMO.JTUser.RRU.Avg	Average number of working RRUs for D-MIMO JT UEs in the downlink

Monitoring the Working Status and Channel Calibration Status of the D-MIMO Cluster

The external CHR counter **PERIOD_CLUSTER_DMIMO_MR** is used to monitor the working status and the channel calibration status of the D-MIMO cluster.

This counter periodically measures the invalidity duration of the D-MIMO cluster. The working status of the D-MIMO cluster can be determined based on the invalidity duration. The default measurement period is 15 minutes.

Field	Description
>DmimoClusterID	ID of the D-MIMO cluster to which the logical cells performing MR reporting belong
>ClusterAbnormallatency	Invalidity duration of the D-MIMO cluster
>CalibrationAbnormallatency	Channel calibration invalidity duration of the D-MIMO cluster

Monitoring the Pairing Status in a D-MIMO Cell

On the MAE-Access, the following **MIMO (Cell) Monitoring** items can be used to monitor the pairing status in a D-MIMO cell.

Monitoring Item	Unit	Description
Number of Enable DMIMO Pairing RB	Number	Number of RBs that can be paired for D-MIMO in a cell within a monitoring period. The total number of available RBs that can be paired for D-MIMO during all transmission time intervals (TTIs) is averaged at the end of the monitoring period.
Number of successful DMIMO Pairing TM7 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM7 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$
Number of successful DMIMO Pairing TM8 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM8 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$

Monitoring Item	Unit	Description
Number of successful DMIMO Pairing TM9 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM9 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$

Monitoring the Pairing Status of a Single UE in a D-MIMO Cell

On the MAE-Access, the following **MIMO(User) Monitoring** items can be used to monitor the pairing status of a single UE in a D-MIMO cell.

Monitoring Item	Unit	Description
Number of Enable DMIMO Pairing RB(Num)	Number	Number of RBs that can be paired for a D-MIMO UE within a monitoring period. The total number of available RBs that can be paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. This item is displayed only when the Test Items parameter is set to DMIMO .
Number of successful DMIMO pairing RB (Num) with N Layers	Number	Number of RBs for UEs successfully paired for D-MIMO at layer N within a monitoring period. The total number of RBs for UEs successfully paired for D-MIMO at layer N during all TTIs is averaged as a cell-level value (including all RRUs) at the end of a monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$ This item is displayed only when the Test Items parameter is set to DMIMO . Value range: 0–1000

Monitoring CEU Performance

The following counters can be used to monitor CEU performance on the eNodeB side.

Counter Name	Counter Description
L.Thrp.bits.UL.BorderUE.JointReception	Total uplink PDCP-layer traffic volume jointly received for CEUs in a cell
L.Thrp.bits.UL.SmallPkt.BorderUE.JointReception	PDCP-layer traffic volume scheduled for uplink small packets jointly received for CEUs in a cell
L.Thrp.Time.UL.RmvSmallPkt.BorderUE.JointReception	Transmission duration of uplink data except small packets jointly received for CEUs in a cell
L.Thrp.Time.UL.BorderUE.JointReception	Total duration of jointly receiving data from CEUs in the uplink at the PDCP layer in a cell
L.Thrp.bits.DL.BorderUE.JointTransmit	Total downlink PDCP-layer traffic volume jointly transmitted for CEUs in a cell
L.Thrp.bits.DL.LastTTI.BorderUE.JointTransmit	Downlink PDCP traffic volume jointly transmitted for CEUs in the last TTI before the buffer is empty
L.Thrp.Time.DL.RmvLastTTI.BorderUE.JointTransmit	Data transmission duration for joint-transmission CEUs except the last TTI before the downlink buffer is empty
L.Thrp.Time.DL.BorderUE.JointTransmit	Total duration of jointly transmitting data to CEUs in the downlink at the PDCP layer in a cell
L.ChMeas.PRB.PUSCH.Avg.BorderUE.JointReception	Average number of PUSCH PRBs used by CEUs for joint reception in a cell
L.ChMeas.PRB.PDSCH.Avg.BorderUE.JointTransmit	Average number of PDSCH PRBs used by CEUs for joint transmission in a cell

4.4.4 Possible Issues

D-MIMO cells automatically revert to SFN cells if there is a fault related to channel calibration, activation of a cell in a D-MIMO cluster, or RRUs.

Channel Calibration Failures

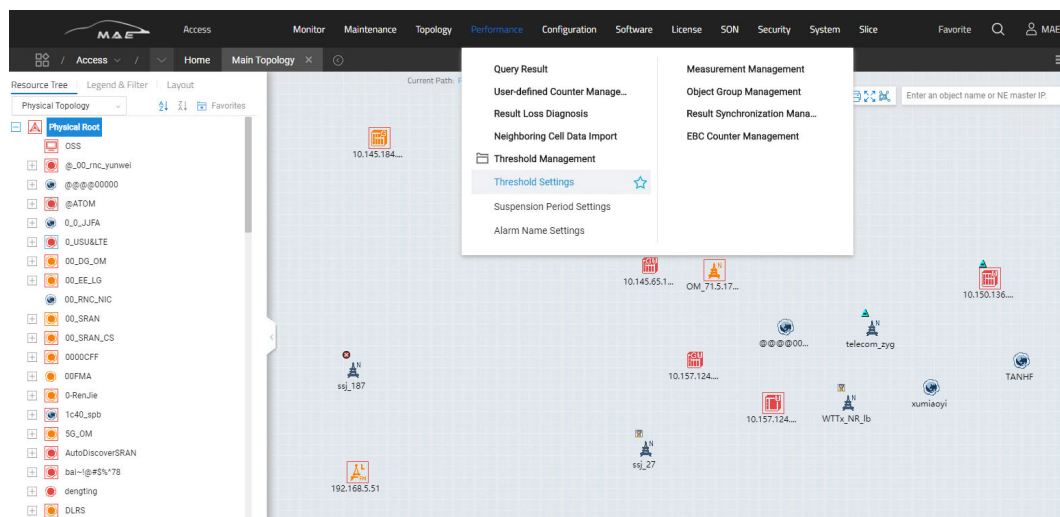
A user-defined performance threshold alarm can be configured on the MAE-Access. The alarm is reported when the duration of the channel calibration failure exceeds the configured threshold. The alarm helps O&M personnel more quickly identify channel calibration failures.

NOTE

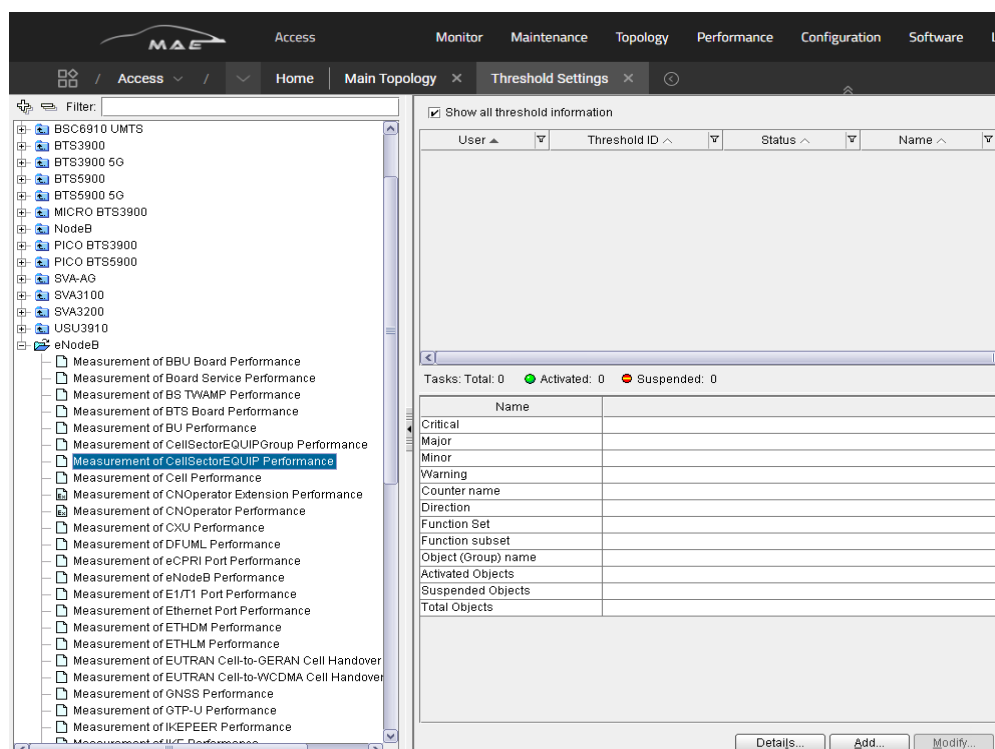
The **L.CellSectorEqpt.UNA.Dur.Cali** counter measures the channel calibration failure duration.

To define a channel calibration failure alarm, perform the following operations on the MAE-Access:

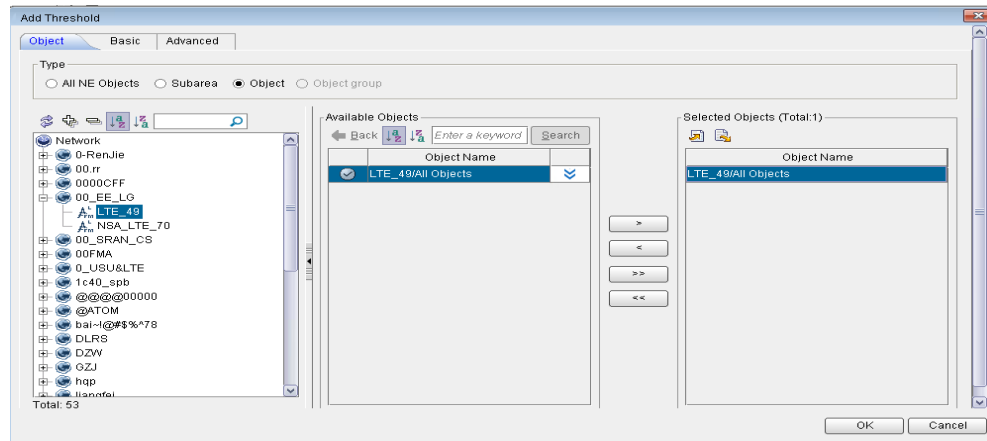
Step 1 On the MAE-Access, choose **Performance > Threshold Management > Threshold Settings**. The **Threshold Settings** tab page is displayed.



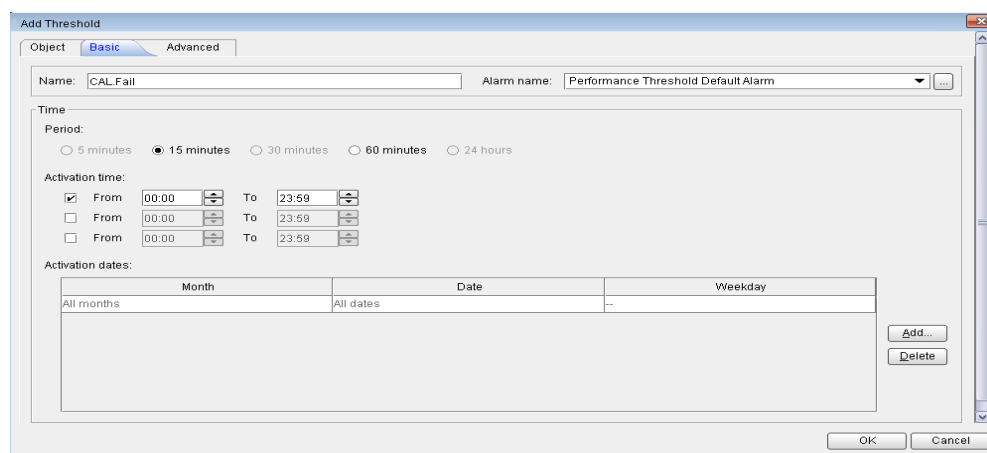
Step 2 Choose **eNodeB > Measurement of CellSectorEQUIP Performance** from the navigation tree, and then click **Add** at the bottom right part of the tab page. The **Add Threshold** dialog box is displayed.



Step 3 Select some or all NEs in the **Object** tab of the **Add Threshold** dialog box.



Step 4 On the **Basic** tab page, set **Name**, **Period**, and **Activation time**.



Step 5 On the **Advanced** tab page, perform the following operations and then click **OK**.

An alarm is reported if the duration of the channel calibration failure is greater than or equal to the threshold plus the offset. This alarm is cleared if the duration falls below the threshold minus the offset.

- Set **Direction** to **Ascending**.
- Set **Function subset** to **Network / Measurements Related to Algorithm(LTE) / Cell Sector Algorithm Measurement**.
- Set **Counter name** to **L.CellSectorEqpt.UNA.Dur.Cali**.
- Set **Threshold** based on the calculation result of **Period** set on the **Basic** tab page multiplied by 80% minus the offset. In the example, set **Threshold** to **9**, namely $15 \times 80\% - 3$.
- Set **Offset** based on the calculation result of **Period** set on the **Basic** tab page multiplied by 20%. In the example, set **Offset** to **3**, namely $15 \times 20\%$.

----End

Channel Calibration-related Troubleshooting

If no user-defined performance threshold alarms are reported on the MAE-Access, run the **DSP DMIMOCALIBRATION** command to query the channel calibration status. In the command output, if the value of the **D-MIMO Calibration Result** parameter is not **Succeeded**, there are the following other possible values and actions to be taken:

- If the value is **Exception occurred in internal calibration**, intra-RRU channel calibration has failed. Run the **DSP CELLCALIBRATION** command to identify the faulty RRU and then contact Huawei technical support for fault handling.
- If the value is **Route search failed**, no appropriate routing path has been found between RRUs because the RRU experiences external interference. The interference needs to be eliminated first.
- If the value is **Reciprocity calibration failed**, the intra-cluster RRU route search was successful but inter-RRU reciprocity calibration failed due to the failure to reach specified thresholds. Obtain the values of the **Calibration Signal CINR** and **Calibration Signal RSSI** parameters and then contact Huawei technical support for troubleshooting.

Faults Related to Activation of Cells in a D-MIMO Cluster

The value of the **D-MIMO Cluster Cell Status** parameter is not **Normal** in the **DSP DMIMOCLUSTERCELL** command output.

- If the value is **Incorrect configurations or unavailable licenses**, check whether the following conditions are met: (1) The D-MIMO switch is turned on. (2) The number of RRUs is configured properly. (3) The D-MIMO cluster is configured properly. (4) The license has been activated.
- If the value is **Limited hardware capacity**, check whether the cell is set up on a UBBPd/UBBPc.
- If the value is **Route application failures**, check whether the routing bandwidth is sufficient.
- If the value is **Cell abnormal**, run the **DSP CELL** command to verify the cell status.

- If the value is **Insufficient co-processing resources**, check whether co-processing board configurations meet the requirements provided in [4.3.3 Hardware](#).
- If the value is **Co-processing resources abnormal**, check whether co-processing board configurations meet the requirements provided in [4.3.3 Hardware](#).
- If the value is **Channel calibration failures**, run the **DSP DMIMOCALIBRATION** command to identify the cause.
- If the value is **Cluster ID conflicts**, check whether the cluster ID is configured according to the restraints provided in [4.3.4 Networking](#).
- If the value is **Clusters being established**, wait until the cluster is set up successfully.

Others

When the **DmimoJTSwitch** option of the **CellAlgoSwitch.DMIMOAlgoSwitch** parameter is selected, consecutive block errors may arise during dual-stream beamforming, even leading to network disconnections. For details about the workarounds, see *Beamforming (TDD)*.

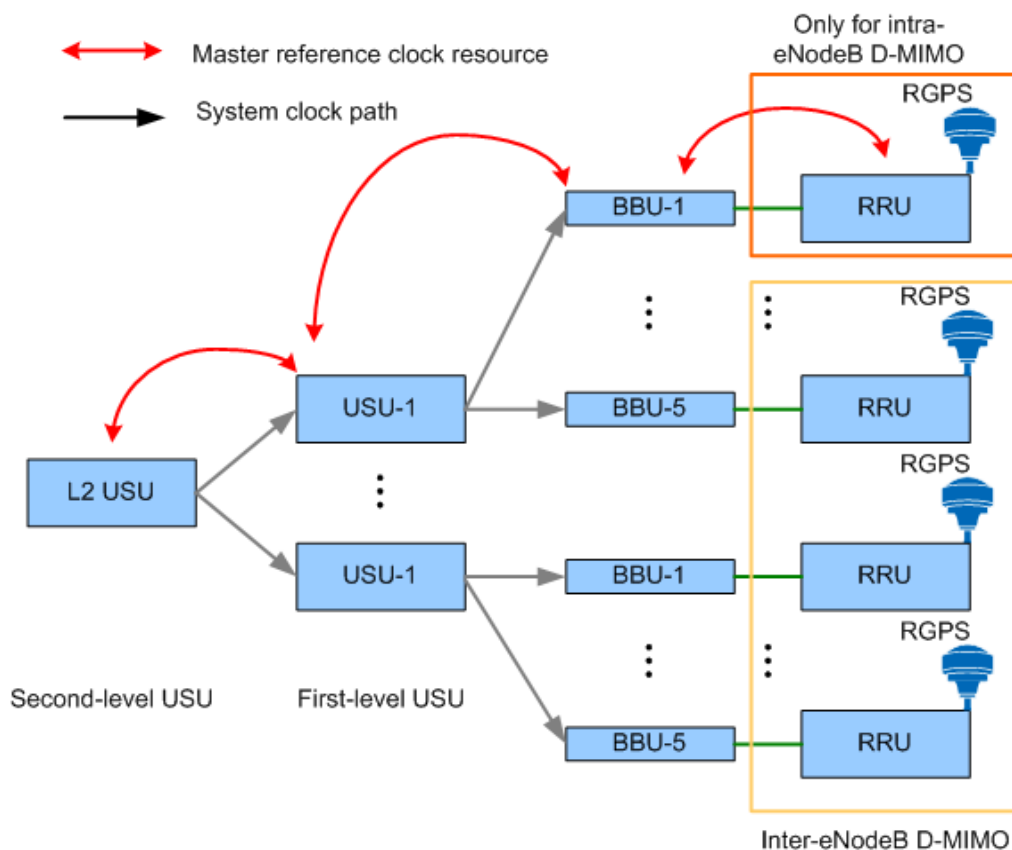
5 Inter-eNodeB D-MIMO

5.1 Principles

In this networking, cells generated by the RRUs connected to different BBUs but on the same Cloud BB network to form a D-MIMO cluster. Inter-eNodeB downlink D-MIMO uses inter-eNodeB inter-RRU channel calibration and coherent joint transmission. For details about inter-RRU channel calibration and coherent joint transmission, see [3 General Principles](#).

[Figure 5-1](#) shows inter-eNodeB D-MIMO networking. This scenario allows for inter-BBU RRU networking and requires full frequency synchronization across these BBUs. If RGPS functions as the clock source of the entire Cloud BB network, the BBU on which the RGPS device is configured is available only for intra-eNodeB D-MIMO services. This restriction does not apply when a GPS clock functions as the clock source of the entire Cloud BB network.

Figure 5-1 Networking diagram of inter-eNodeB D-MIMO



The inter-BBU clock solution is implemented as follows:

1. BBU 1 connects to the clock reference source (for example, RGPS), locks the reference clock source, and shares the clock source with the USUs.
2. The USUs provide a system clock source.
 - If two levels of USUs are configured, the second-level USU is selected as the root node to provide the system clock.
 - If only one level of USUs is configured, the first-level USU is selected as the root node to provide the system clock.
3. The hardware synchronization modules of other BBUs are locked to the system clock source generated in step 2.

5.2 Network Analysis

5.2.1 Benefits

This feature offers the same benefits as downlink D-MIMO. For details, see [4.2.1 Benefits](#).

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

The network impact of this function is the same as that of downlink D-MIMO. For details, see [4.2.2 Impacts](#).

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
TDLEOFD-121501	Inter-eNodeB DL D-MIMO (Trial)	LT1STDLDMC00	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Downlink D-MIMO	DmimoJTSwitch option of the CellAlgoSwitch.DMIMOAlgoSwitch parameter	<i>D-MIMO (TDD)</i>	Inter-eNodeB DL D-MIMO depends on downlink D-MIMO.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations are compatible with this feature.

Boards

For details about the requirements for BBPs and co-processing boards, see [Boards](#). This feature also has the following constraints:

- For the main control board:
The main control board must be a UMPTe/UMPTg/UMPTe+UMPTg/.
- The D-MIMO cluster and co-processing board must be configured on the BBU where layer 2 of the SFN cell is configured in inter-eNodeB D-MIMO scenarios.

RF Modules

This feature depends on downlink D-MIMO. For details, see [RF Modules](#).

Cells

This feature depends on downlink D-MIMO. For details, see [Cells](#).

5.3.4 Networking

This feature requires downlink D-MIMO. For details about the networking requirements of downlink D-MIMO, see [4.3.4 Networking](#).

Planning of D-MIMO Cluster IDs

D-MIMO cluster IDs need to be planned using a D-MIMO cluster planning tool, which can be obtained from Huawei engineers.

D-MIMO cluster planning must meet the following requirements:

- Basic requirements
 - Cells generated by more than one RRU or more than two pRRUs form a D-MIMO cluster.
 - Each cluster ID must have a unique mod 6 value for each RRU in multi-carrier scenarios. The value of the **D-MIMO Calibration Result** parameter is **Channel calibration failures** in inter-eNodeB scenarios if multiple cluster IDs corresponding to multiple carriers of the same RRU have the same mod 6 value.
 - The D-MIMO cluster ID must be unique within a BBU.
 - The IDs of physically neighboring D-MIMO clusters must not have the same mod 24 value.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Parameters Used for Activation

Inter-eNodeB D-MIMO is deployed by performing the following operations:

- Enable downlink D-MIMO. For details, see [4.4.1.1 Data Preparation](#).
- Set the following clock parameters for inter-eNodeB services.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	TASM.MODE	Set this parameter for the USU according to the following principles: • This parameter must not be set to FREE for top-level USUs. • This parameter can be set to FREE for lower-level USUs.

Parameter Name	Parameter ID	Setting Notes
Locked System Clock Source	TASM.SYSCLKSRC	Set this parameter according to the following principles when one level of USU is configured: • Set this parameter to LOCAL for first-level USUs. • Set the TASM.SYSCLKSRC parameter to INTER_SYSCLK for the BBU. Set this parameter according to the following principles when two levels of USUs are configured: • Set this parameter to LOCAL for the second-level USU. • Set this parameter to INTER_SYSCLK for first-level USUs. • Set the TASM.SYSCLKSRC parameter to INTER_SYSCLK for the BBU.

The following requirements also need to be satisfied during inter-eNodeB D-MIMO deployment:

- If RGPS functions as the clock source for the entire Cloud BB network, the BBU on which the RGPS device is configured cannot be configured in an inter-eNodeB D-MIMO cluster, but is available only for intra-eNodeB D-MIMO services, not for inter-eNodeB D-MIMO services. This restriction does not apply when a GPS clock functions as the clock source of the entire Cloud BB network.
- Inter-eNodeB D-MIMO is not supported when there is a backup for the first-level USU or second-level USU.

5.4.1.2 Using MML Commands

Inter-eNodeB D-MIMO is deployed by performing the following operations:

- Enable downlink D-MIMO. For details, see [4.4.1.2 Using MML Commands](#).
- Set clock parameters for inter-eNodeB services. For details, see *USU3910-based Multi-BBU Interconnection*.

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

This feature requires downlink D-MIMO. It has the same activation observation methods as downlink D-MIMO. For details, see [4.4.2 Activation Observation](#). If the **DSP DMIMOCALIBRATION** command output shows that the route search fails, the **DSP INREC** command cannot be executed on the secondary BBU to query the inter-RRU calibration interference power. In other cases, the **DSP INREC** command can be executed on the secondary BBU to query the inter-RRU calibration interference power.

5.4.3 Network Monitoring

This feature requires downlink D-MIMO. It has the same network monitoring methods as downlink D-MIMO. For details, see [4.4.3 Network Monitoring](#).

5.4.4 Possible Issues

The value of the **D-MIMO Cluster Cell Status** parameter is not **Normal** in the **DSP DMIMOCLUSTERCELL** command output.

- If the value is **Clock exceptions**, run the **DSP SYSCLKSRC** command to check whether the value of the **Locked System Clock Source** parameter is **Interconnection System Clock** and the value of the **ESN of the NE where the locked source is located** parameter is the same for all BBUs and USUs. If not, check clock configurations.
- If the value is **Incorrect configurations or unavailable licenses**, check whether the inter-eNodeB link is functional.

If there are other faults, see [4.4.4 Possible Issues](#).

6 Inter-Cell D-MIMO

6.1 Principles

Inter-cell D-MIMO enables RRUs with overlapping coverage and corresponding physical cells under the same BBU to form a D-MIMO cluster. There are multiple PCs within the D-MIMO cluster.

- In outdoor macro-macro scenarios, an inter-cell D-MIMO cluster only allows for common cells, not a combination of common cells and SFN cells.
- In LampSite scenarios, inter-cell D-MIMO is compatible with SFN. However, each pRRU must serve only one cell.

There are two types of inter-cell D-MIMO:

- Intra-BBP inter-cell D-MIMO
Layers 1 and 2 of the cells in the D-MIMO cluster share the same BBP, as shown in [Figure 6-1](#).
- Inter-BBP inter-cell D-MIMO
Cells in a D-MIMO cluster are set up on different BBPs housed in the same BBU, as shown in [Figure 6-2](#).

Figure 6-1 Intra-BBP inter-cell D-MIMO

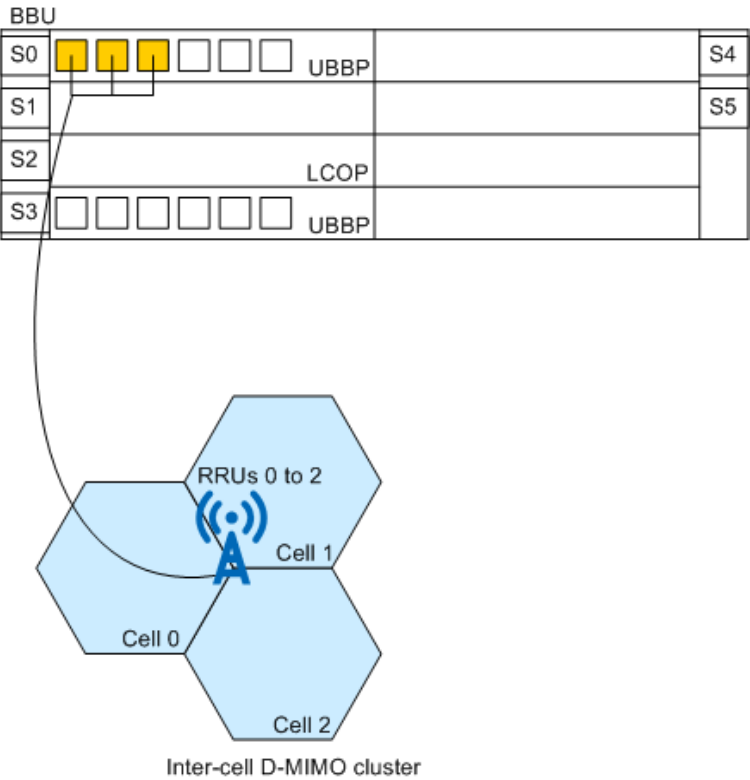
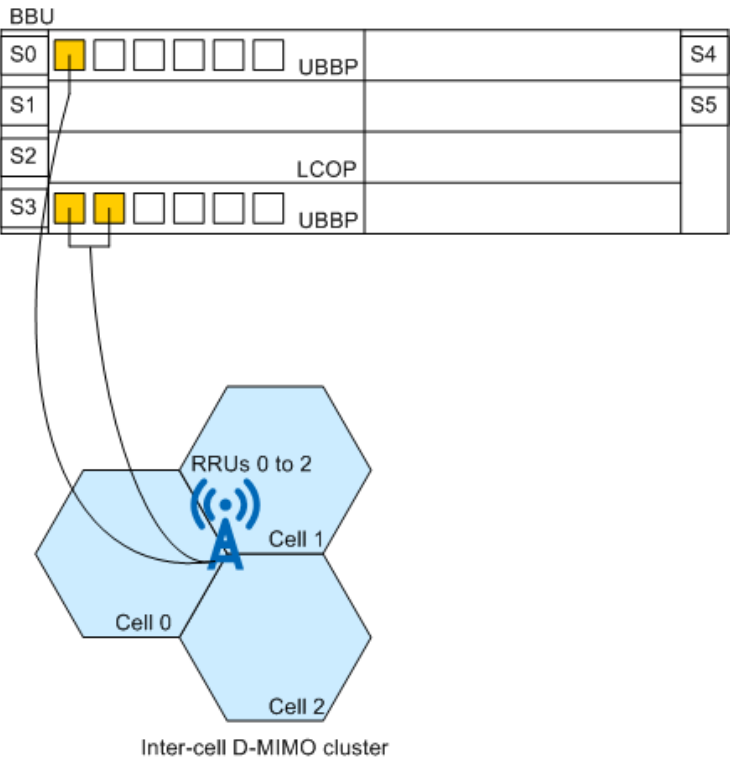


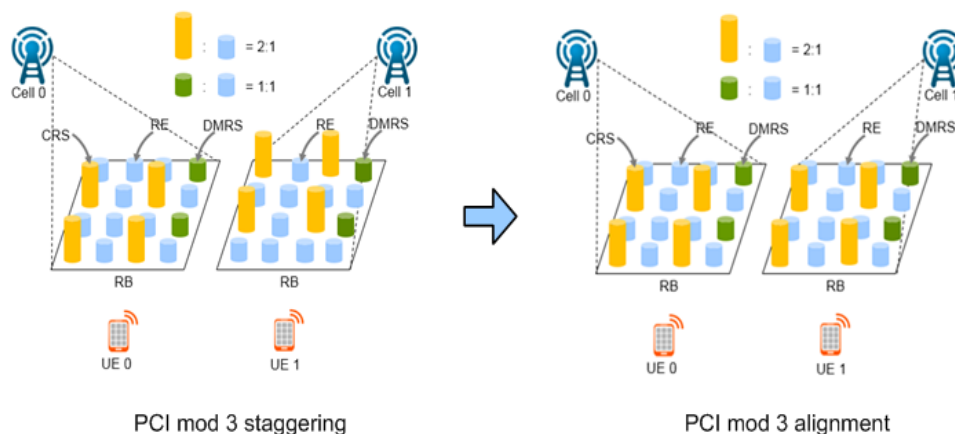
Figure 6-2 Inter-BBP inter-cell D-MIMO



To reduce inter-cell interference in the D-MIMO cluster, PCI mod 3 values need to be aligned for cells in the cluster. [Figure 6-3](#) shows the PCI mod 3 alignment

principles. If PCI mod 3 values are not aligned, the CRSs of neighboring cells face data REs of the local cell, which creates interference. By aligning PCI mod 3 values, CRSs and data REs are aligned separately between neighboring cells and the local cell, maximizing the JT gains.

Figure 6-3 Network reconstruction for PCI mod 3 alignment



Inter-cell D-MIMO is suitable in the following scenarios:

- Outdoor macro-macro scenarios
- LampSite scenarios

Inter-cell D-MIMO is suitable for outdoor hotspot areas, such as densely populated urban areas, central business districts (CBDs), and campuses. An inter-cell D-MIMO cluster has resources of multiple logical cells and therefore can serve more UEs than intra-cell D-MIMO.

NOTE

"Macro-macro networking" mentioned in this document refers to the combination of macro RRUs. For details, see [RF Modules](#).

Related Optimization

The coordinated scheduling experience optimization function has been added in eRAN TDD 13.1. This function is controlled by the **DL_COORD_SCH_EXP_OPT_SW** option of the **CellAlgoSwitch.DLSchExtSwitch** parameter. The proportion of coordinated scheduling increases and the CEU perceived rate improves when this function is used with inter-cell D-MIMO. This function is recommended when the CEU perceived rate is low.

6.2 Network Analysis

6.2.1 Benefits

Most Beneficial Scenarios

Inter-cell D-MIMO is suitable when there is significant overlap between cells, the UE speed is less than 5 km/h, and the inter-site distance is short (300 m or less for

outdoor scenarios and LOS between the pRRUs in LampSite scenarios). This function has the following requirements:

- In macro-macro networking scenarios, more than 20% of the intra-frequency coverage area within the 6 dB scope overlap, the PRB usage is greater than 10%, and the proportion of beamforming UEs in macro cells is greater than 20%.

Inter-cell D-MIMO is not recommended in either of the following scenarios:

- High-speed scenarios
Inter-cell D-MIMO is not suitable in high-speed scenarios, because D-MIMO depends on beamforming.
- Scenarios with too many UEs
In most cases, only UEs allocated with short SRS periods can enter beamforming. If there are too many UEs in a cell, for example, more than 400 UEs, only a few of them can use beamforming. Therefore, inter-cell D-MIMO is not recommended in such scenarios.
- Scenarios with a low coverage overlap rate

Benefits

Inter-cell D-MIMO provides the following benefits:

- In typical outdoor macro-macro or LampSite networking scenarios, inter-cell D-MIMO increases the average downlink user-perceived rate and the downlink CEU-perceived rate.
- D-MIMO increases the CEU throughput in full-buffer service tests when the coordinating cell meets either of the following conditions:
 - The coordinating cell has remaining PRBs for SU-JT.
 - The coordinating cell has no remaining PRB resources but meets the MU-JT triggering conditions.

If neither of the preceding conditions is met, the CEU throughput is the same as that achieved in normal scenarios.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

- Inter-cell D-MIMO has the following impacts on the reported 4-bit CQI:
The reported 4-bit CQI value increases or decreases depending on the following conditions:
 - The reported 4-bit CQI value decreases in light-load macro-macro scenarios due to a decrease in the CRS SINR after CRS alignment.
 - The reported 4-bit CQI value increases if the proportion of dual-stream beamforming grows in macro-macro scenarios. This is because UEs working in TM3 rank 2 report CQIs for both data streams while UEs working in dual-stream beamforming (TM8) report the CQI for only one data stream.
- Inter-cell D-MIMO slightly decreases the RAR success rate, RRC connection setup success rate, RRC connection reestablishment success rate, and cell

handover success rate, and slightly increases the UE access delay. It increases the average downlink VoLTE delay and call connection delay by less than 10 ms.

- According to section 7.3 in 3GPP TS36.213, the PUSCH carries 4-bit downlink HARQ feedback. When the HARQ feedback mode is multiplexing, the UE does not perform DTX feedback, and missing PDCCH detection converges relatively slowly. In the lab test scenario where full-buffer services are used and downlink HARQ feedback is carried on the PUSCH, the impact of MU-JT in inter-cell D-MIMO on the downlink cell throughput is greater than that of common scheduling. On live networks, there are no full-buffer services and the PUSCH carries downlink HARQ feedback. Therefore, inter-cell D-MIMO is not affected.

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
TDLEOFD-130501	Inter-Cell DL D-MIMO	LT1SICDLDM00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Single-stream beamforming	BfSwitch option of the CellAlgoSwitch.BfAlgoSwitch parameter	<i>Beamforming (TDD)</i>	• In outdoor macro-macro scenarios, single-stream beamforming is a prerequisite function. • In LampSite scenarios, single-stream beamforming is not a prerequisite function and therefore the corresponding license is not required. However, the BfSwitch option of the CellAlgoSwitch.BfAlgoSwitch parameter must be selected.
MU beamforming	MuBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter	<i>Beamforming (TDD)</i>	• In outdoor macro-macro scenarios, MU beamforming is a prerequisite function. • In LampSite scenarios, MU beamforming is not a prerequisite function and therefore the corresponding license is not required. However, the MuBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter must be selected.

6.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Extended CP	• Cell.UlCyclicPrefix • Cell.DlCyclicPrefix	<i>Extended CP</i>	Inter-cell D-MIMO does not support extended CP.
High speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>	Inter-cell D-MIMO does not apply to high-speed scenarios.
Out-of-band relay	CellAlgoSwitch.RelaySwitch	<i>Relay</i>	Inter-cell D-MIMO does not apply to out-of-band relay scenarios.

Function Name	Function Switch	Reference	Description
Massive MIMO introduction	Service-based dynamic SRS allocation is controlled by the SrvBasedSRSAdjAlgo option of the SRSCfg.SrsCfgPolicySwitch parameter. Other functions are free from switch control.	<i>Massive MIMO Basics (TDD)</i>	Inter-cell D-MIMO does not apply to massive MIMO scenarios.
Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD)</i>	When an intra-frequency soft split cell is added to a D-MIMO cluster, inter-cell downlink D-MIMO cannot take effect. Inter-frequency soft split is compatible with inter-cell downlink D-MIMO.
SFN	None	<i>SFN</i>	In inter-cell D-MIMO scenarios, only common cells, not a combination of common cells and SFN cells, can be added to a D-MIMO cluster. In LampSite scenarios, inter-cell D-MIMO is compatible with SFN. However, each pRRU must serve only one logical cell.
Uplink SU-MIMO	CellAlgoSwitch.UlSuMimoAlgoSwitch	<i>MIMO</i>	Inter-cell D-MIMO cannot take effect when uplink SU-MIMO is enabled.
Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	Dynamic Power Sharing Between LTE Carriers cannot be enabled together with inter-cell D-MIMO.
Turbo beamforming	TurboBfSwitch option of the CellAlgoSwitch.MuBfAlgoSwitch parameter	<i>WTTx Turbo Beamforming (TDD)</i>	Turbo beamforming cannot be enabled together with inter-cell D-MIMO.

Function Name	Function Switch	Reference	Description
Enhanced coordinated scheduling based power control	CellCspcPara.CelleCspcSwitch	<i>Massive MIMO Enhancements (TDD)</i>	Enhanced coordinated scheduling based power control cannot be enabled together with inter-cell D-MIMO.
SRS interference avoidance	WTTxSRSIntrfAvoidanceSw option of the SRSCfg.SrsCfgPolicySwitch parameter	<i>Massive MIMO Optimization in WTTx Scenarios (TDD)</i>	SRS interference avoidance cannot be enabled together with inter-cell D-MIMO.
Energy saving based on proactive scheduling	SymbolPwrSaving.TrigBndlSchDlvPrbThld	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on proactive scheduling cannot be enabled together with inter-cell D-MIMO.
RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None
Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and rank-4 closed-loop TM9	TM3R2_TO_TM9R4_THLD_ADAPT_SW option of the CellBfMimoParaCfg.BfMimoAlgoOptSwitch parameter	<i>TMA (TDD)</i>	Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and rank-4 closed-loop TM9 does not take effect if the InterCellDmimoJTSwitch option of the CellAlgoSwitch.DMIMOAlgoSwitch parameter is selected.

Function Name	Function Switch	Reference	Description
Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and dual-stream beamforming	TM3R2_TO_DUAL_BF_THLD_ADAPT_SW option of the CellBfMimoParaCfg.BfMimoAlgoOptSwitch parameter	<i>TMA (TDD)</i>	Dynamic cell-level adjustment of the thresholds for switching between rank-2 TM3 and dual-stream beamforming does not take effect if the InterCellDmimoJTSwitch option of the CellAlgoSwitch.DMIMOAlgoSwitch parameter is selected.
Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>	None
Adaptive MU beamforming	CellBf.AdaptMubfStartThld set to a value other than 255	<i>Beamforming (TDD)</i>	None
MuteUppTsSymbForSrsSw	MuteUpptsSymbForSrsSw option of the SRSCfg.SrsCfgPolicySwitch parameter	None	The switch specified by the MuteUpptsSymbForSrsSw option cannot be turned on together with the switch for inter-cell D-MIMO.
Single-cell intelligent power control energy saving	INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None

6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
DRX	CellDrxPara.DrxAlgoSwitch	<i>DRX and Signaling Control</i>	When DRX is enabled, UEs do not transmit the SRS during DRX sleep time. In such cases, the instantaneous beamforming weighting values of D-MIMO UEs cannot be updated promptly, causing D-MIMO performance to deteriorate.
Adaptive switching between beamforming and MIMO	CellBfMimoParaCfg.BfMimoAdaptiveSwitch	<i>TMA (TDD)</i>	• Light-load scenarios TMA oriented for optimal single-link performance is used. • Heavy-load scenarios To increase the cell capacity, TMA oriented for optimal single-link performance cannot be used. Cell-level TMA is recommended when the cell load is heavy and multiple layers are available for pairing.

Function Name	Function Switch	Reference	Description
DL CoMP	CellAlgoSwitch.DLCompSwitch	<i>DL CoMP (TDD)</i>	When inter-cell D-MIMO is enabled, D-MIMO is used in interference suppression for CEUs and the performance counters related to TDLAOFD-003002 Intra-eNodeB DL CoMP in Adaptive Mode are measured in a D-MIMO way. • The reserved SRS period can be configured. – The SRSCfg.RsvResSrsPeriod1 parameter specifies the SRS period for resources reserved in the first half frame. – The SRSCfg.RsvResSrsPeriod2 parameter specifies the SRS period for resources reserved in the second half frame. • The SRSCfg.RsvResSrsPeriod1 and SRSCfg.RsvResSrsPeriod2 parameter settings must be the same for cells belonging to the same CoMP cluster. Intra-eNodeB DL CoMP does not take effect in the cells where inter-cell D-MIMO has taken effect. If intra-eNodeB DL CoMP takes effect first and then inter-cell D-MIMO takes effect in a cell, the GUI shows that both of them take effect within no more than one minute but actually only inter-cell D-MIMO takes effect. In this period, intra-eNodeB DL CoMP traffic statistics are not collected and related items are not monitored on the MAE-Access.
Downlink frequency selective scheduling	FreqSelSwitch option of the CellAlgoSwitch.DLSchSwitch parameter	<i>Downlink Scheduling</i>	When both inter-cell D-MIMO and downlink frequency selective scheduling are enabled, D-MIMO joint transmission is preferentially performed.

Function Name	Function Switch	Reference	Description
Uplink coordinated AMC	ULCamcSw option of the CellAlgoSwitch.CamcSwitch parameter	<i>Uplink Coordinated Scheduling</i>	- The reserved SRS period can be configured. - The SRSCfg.RsvResSrsPeriod1 parameter specifies the SRS period for resources reserved in the first half frame. - The SRSCfg.RsvResSrsPeriod2 parameter specifies the SRS period for resources reserved in the second half frame. - The SRSCfg.RsvResSrsPeriod1 and SRSCfg.RsvResSrsPeriod2 parameter settings must be the same for cells belonging to the same uplink coordinated AMC cluster.
Uplink inter-eNodeB coordinated AMC	ULInterEnbCamcSw option of the CellAlgoSwitch.CamcSwitch parameter	<i>Uplink Coordinated Scheduling</i>	
Downlink CA	None	<i>Carrier Aggregation</i>	When both inter-cell D-MIMO and downlink CA are enabled, the SCells of a UE cannot be selected for D-MIMO joint transmission for the UE.
RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When a cell in an inter-cell D-MIMO cluster enters the RF channel intelligent shutdown state, inter-RRU channel calibration fails, causing the inter-cell D-MIMO cluster to be disabled.
Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When a cell in an inter-cell D-MIMO cluster enters the intelligent carrier shutdown state, inter-RRU channel calibration fails, causing the inter-cell D-MIMO cluster to be disabled.
Low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	When a cell in an inter-cell D-MIMO cluster is in low power consumption mode, it cannot serve as a cooperating cell for joint transmission and inter-RRU channel calibration may fail, causing the inter-cell D-MIMO cluster to be disabled.

Function Name	Function Switch	Reference	Description
Downlink 2-layer MIMO based on TM9	TM9Switch option of the CellAlgoSwitch.EnhMIMOSwitch parameter	<i>Beamforming (TDD)</i>	When TM9 is enabled and the CSI-RS is configured, joint transmission is performed only if all cells within an inter-cell D-MIMO cluster have the same CSI-RS time-frequency position.
Downlink 4-layer MIMO based on TM9			
RAN sharing with common carrier	ENodeBSharingMode.EnodeBSharingMode	<i>RAN Sharing</i>	The cells in an inter-cell D-MIMO cluster must belong to the same operator. In these cells, joint transmission can be used only for UEs served by this operator in the D-MIMO cluster. In inter-cell D-MIMO scenarios, joint transmission by multiple cells can be performed only for the UEs served by the primary operator.
Adaptation between beamforming and MIMO with TM4	CellBfMimoParaCfg.BfMimoAdaptWithTm4Switch	<i>TMA (TDD)</i>	Joint transmission cannot be performed for UEs in TM3 or TM4.
Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	When a cell in an inter-cell D-MIMO cluster enters the intelligent carrier shutdown state, inter-RRU channel calibration fails, causing the inter-cell D-MIMO cluster to be disabled.
Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MUL_CARRIER_HIER_MANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	When a cell in an inter-cell D-MIMO cluster enters the multi-carrier coordinated energy saving state, inter-RRU channel calibration fails, causing the inter-cell D-MIMO cluster to be disabled.

Function Name	Function Switch	Reference	Description
SOC level-based low power consumption mode	IPWRM.SOCLVLA DJSW SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	When a cell in an inter-cell D-MIMO cluster enters SOC level-based low power consumption mode, it cannot serve as a coordinating cell for joint transmission and inter-RRU channel calibration may fail, causing the inter-cell D-MIMO cluster to be disabled.
Downlink FDD+TDD CA	InterFddTddCaSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-cell downlink D-MIMO does not work in SCells.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The following base stations are compatible with this feature:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

NOTE

For details about the available bandwidths and configuration restraints in LampSite scenarios, see *LampSite Solution Description*, *LampSite BBU Technical Specifications*, *LampSite pRRU&RRU Technical Specifications*, and *LampSite RHUB Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

Boards

- Baseband processing unit (BBP)
The UBBPg, UBBPd6, UBBPd9, or UBBPe series board is required for setting up cells and connecting to RRUs corresponding to the cells through optical fibers.
The UBRlb board does not support D-MIMO.
An FDD+TDD UBBPg board is incompatible with inter-cell D-MIMO.
In inter-cell D-MIMO scenarios, all cells in a D-MIMO cluster can be set up on different BBPs but must be in the same BBU.
The outdoor BBU3910A does not support D-MIMO or enhanced channel calibration.

- Dedicated cooperative processing board
This feature requires an LTE cooperative processing board (LCOP) to calculate inter-cell D-MIMO co-processing resources. Each cooperative processing board supports a maximum of 288 antennas, 36 cells, and 12 D-MIMO clusters.
The LCOP uses the base station frame offset specified by the **ENodeBFrameOffset.TddFrameOffset** parameter. To ensure cell coordination in a cluster, the cell frame offset specified by the **CellFrameOffset.FrameOffset** parameter for each cell in the cluster needs to be the same as the base station frame offset.

RF Modules

NOTE

When the RRU3279 serves two 4T4R sectors by splitting, it does not support D-MIMO.
NR-capable RF modules do not support D-MIMO.

Category	RRU Model	Number of Channels	Supported Frequency Band	Scenario
Macro RRU	RRU3252	4T4R	Band 38	Outdoor macro-macro
Macro RRU	RRU3256	4T4R	Band 42 or band 43	Outdoor macro-macro
Macro RRU	RRU3278	8T8R	Band 42 or band 43	Outdoor macro-macro
Macro RRU	RRU3279	8T8R	Band 38	Outdoor macro-macro
pRRU	pRRU3901	2T2R	2.3 GHz	Indoor LampSite
pRRU	pRRU3911	2T2R	2.3 GHz, 2.5 GHz, or 2.6 GHz	Indoor LampSite

In addition, there are the following constraints:

- RRUs cannot be combined.
- Macro RRUs cannot be cascaded.
- RRUs cannot be configured using the load sharing networking.
- RRUs cannot be configured using the ring networking.
- The remote RRU deployment distance cannot exceed 20 km.
- No more than three carriers can be configured on the same RRU.
- In FDD+TDD RRU cascading, the upper-level RRU can be a TDD RRU, not an FDD RRU.

Cells

- The cell bandwidth must be 10 MHz or 20 MHz.
- Bands 38, 40, 42, and 43 are supported in outdoor scenarios.
- The 2.3 GHz and 2.5 GHz bands and band 41 are supported in LampSite scenarios.
- A cell configured with simulated loads cannot function as the serving cell or coordinating cell of a D-MIMO UE.

Configuration Restraints in LampSite Scenarios

- Each pRRU must be configured for a single cell.
- pRRUs belonging to the same D-MIMO cluster must be connected to the same RHUB or connected to two neighboring RHUBs on the same CPRI link.
- No extender is used to extend the Ethernet cable connecting the RHUB to the pRRU.

6.3.4 Networking

Inter-cell D-MIMO clusters need to be planned using a D-MIMO cluster planning tool, which can be obtained from Huawei engineers.

D-MIMO cluster planning must meet the following requirements:

- Basic requirements
 - Each D-MIMO cluster supports at least two cells in macro networking scenarios.
 - Cells belonging to the same D-MIMO cluster must be configured with the same cell bandwidth, frequency, uplink-downlink subframe configuration, special subframe configuration, and the number of CRS ports.
 - Cells in the same D-MIMO cluster must have the same PCI mod 3 value. PCIs of the cells belonging to the D-MIMO cluster and neighboring cells need to be planned using the PCI planning tool, which can be obtained from Huawei engineers.
- Macro-macro scenarios

A D-MIMO cluster allows for up to four macro cells (with a maximum of 32 antennas).
- LampSite scenarios
 - A D-MIMO cluster allows for three to six pRRUs.
 - If multiple carriers are configured, both the number of LTE TDD carriers and the LTE TDD bandwidth must be the same for all the pRRUs belonging to the same D-MIMO cluster, regardless of whether D-MIMO is enabled for all the carriers.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Parameters Used for Activation

The following parameters need to be set when an inter-cell D-MIMO cluster needs to be set up.

Table 6-1 Parameters in the **CoProcRes** MO

Parameter Name	Parameter ID	Setting Notes
Coordinate Process Resource ID	CoProcRes.CoProcResId	Set this parameter to a value ranging from 0 to 12 .
BaseBand Equipment ID	CoProcRes.BaseBandEqmId	Set this parameter to the ID of the baseband equipment on which the dedicated co-processing board is configured. Set this parameter to a value ranging from 0 to 23 .
Work Mode	CoProcRes.WorkMode	Select the COORDINATING_PROCESSING option and deselect the BASEBAND_PROCESSING option if inter-cell D-MIMO is supported.
Bundling Cluster Type	CoProcRes.BundlingClusterType	- Set this parameter to DMIMO if only inter-cell D-MIMO is supported. - Set this parameter only to ADAPTIVE if D-MIMO and other coordination features are supported.

NOTE

- Inter-cell D-MIMO can only support the dedicated co-processing board, which does not support baseband processing. In this case, the **BASEBAND_PROCESSING** option of the **CoProcRes.WorkMode** parameter must not be selected.
- If the **COORDINATING_PROCESSING** option of the **CoProcRes.WorkMode** parameter is selected for a BBP that does not support coordinated processing, ALM-26245 Configuration Data Inconsistency will be generated. In this case, only universal baseband processing of the BBP takes effect so that cell services are not affected.

Table 6-2 Parameters in the **DMIMOCluster** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Cluster ID	DMIMOCluster.DMIMOClusterId	Set this parameter to a value ranging from 0 to 65535. • The D-MIMO cluster ID must be an even number if more than four pRRUs are configured in LampSite scenarios. • Each cluster ID must have a unique mod 6 value for each RRU in multi-carrier scenarios. • The D-MIMO cluster ID must be unique within a BBU.
Coordinate Process Resource ID	DMIMOCluster.CoProcResId	Set this parameter to 255 or to a value ranging from 0 to 12. No co-processing resources are specified when this parameter is set to 255.
Beamforming Weight Normalized Mode	DMIMOCluster.BfWeightNormalizeMode	• If this parameter is set to NEBF, paired beamforming UEs achieve optimal performance when residing at the D-MIMO cell edge. • If this parameter is set to PEBF, paired beamforming UEs achieve optimal performance when residing at the center or at a medium distance from the center of the D-MIMO cell. Set this parameter to NEBF in commercial use scenarios.

Table 6-3 Parameters in the **DMIMOClusterCell** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Cluster ID	DMIMOClusterCell.DMIMOClusterId	Set this parameter to a value ranging from 0 to 65535.

Parameter Name	Parameter ID	Setting Notes
Cell ID	DMIMOClusterCell.CellId	Set this parameter to a value ranging from 0 to 255 .
eNodeB ID	DMIMOClusterCell.eNodeBId	Set this parameter to a value ranging from 0 to 1048575 .
Mobile Country Code	DMIMOClusterCell.Mcc	Set this parameter to a value ranging from 000 to 999 .
Mobile Network Code	DMIMOClusterCell.Mnc	Set this parameter to a value ranging from 00 to 99 or from 000 to 999 .

 NOTE

- An inter-cell D-MIMO cluster does not allow for SFN cells. MPRU aggregation cells are not allowed in outdoor macro-macro scenarios.
- In LampSite scenarios, inter-cell D-MIMO allows for multiple MPRU aggregation cells, but a pRRU is configured for a single cell.
- One cell cannot be added to different D-MIMO clusters.
- One cell can be added to a D-MIMO cluster by the primary operator ID or secondary operator ID but not both. It is recommended that the cell be added to a D-MIMO cluster by the primary operator ID.
- One cell cannot be added to a D-MIMO cluster by both the primary operator ID and the secondary operator ID.

Table 6-4 Parameters in the **CellAlgoSwitch** MO

Parameter Name	Parameter ID	Option	Setting Notes
BF algorithm switch	CellAlgoSwitch.BfAlgoSwitch	BfSwitch	Select this option.
MUBF Algorithm Switch	CellAlgoSwitch.MuBfAlgoSwitch	MuBfSwitch	Select this option.
D-MIMO Algorithm Switch	CellAlgoSwitch.DMIMOAlgoSwitch	InterCellDmimoJTS-switch	Select this option.
Enhanced Channel Calibration Switch	CellAlgoSwitch.EnhChnCalSwitch	-	It is recommended that this switch be turned on when D-MIMO is enabled.
Interference avoidance switch	CellAlgoSwitch.AvoidInterfSwitch	AvoidCrsConflictInterfSw	Select this option when PCI mod 3 values are aligned.

Parameter Name	Parameter ID	Option	Setting Notes
DL Scheduling Extension Switch	CellAlgoSwitch.DLSchExtSwitch	DL_COORD_SCH_EXP_OPT_SW	Select this option when inter-cell D-MIMO is enabled.

Table 6-5 Parameters in the **CellPcAlgo** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO Srs Power Control Offset for Sinr Target	CellPcAlgo.DMSrsPcSinrOffset	It is recommended that this parameter be set to 5 when D-MIMO is enabled. Retain the default value 0 when D-MIMO is disabled.

Table 6-6 Parameters in the **DMIMOAlgo** MO

Parameter Name	Parameter ID	Setting Notes
D-MIMO A3 RSRP Offset	DMIMOAlgo.DmimoA3RsrOffset	Set this parameter to -12 (actual value: -6 dB) when inter-cell D-MIMO is enabled.
D-MIMO JT Cell RSRP Difference Threshold	DMIMOAlgo.DmimoJtCellRsrpDiffThld	Set this parameter to -20 (actual value: -10 dB) when inter-cell D-MIMO is enabled.

Table 6-7 Parameters in the **SRSCfg** MO

Parameter Name	Parameter ID	Setting Notes
Reserved Resource SRS Period 1	SRSCfg.RsvResSrsPeriod1	Set this parameter to 10ms when inter-cell D-MIMO is enabled.
Reserved Resource SRS Period 2	SRSCfg.RsvResSrsPeriod2	Set this parameter to 20ms when inter-cell D-MIMO is enabled.

Table 6-8 Parameters in the **CellCounterParaGroup** MO

Parameter Name	Parameter ID	Option	Setting Notes
Edge User Distinguish A3 Offset	CellCounterParaGroup.EdgeUserA3Offset	-	Set this parameter to -12 when inter-cell D-MIMO is enabled.
Cell Counter Algorithm Switch	CellCounterParaGroup.CellCounterAlgorithmSwitch	BasedA3EdgeUserSwitch	Select this option when inter-cell D-MIMO is enabled.

Table 6-9 Parameters in the **NCellSrsMeasPara** MO

Parameter Name	Parameter ID	Option	Setting Notes
SRS Auto Neighbour Cell Measurement Switch	NCellSrsMeasPara.SrsAutoNCellMeasSwitch	-	Set this parameter to ON.
SRS or DMRS Auto Neighbour Cell Measurement A3 Offset	NCellSrsMeasPara.NCellSrsMeasA3Offset	-	-12
Neighbouring Cell Measurement Switch	NCellSrsMeasPara.NCellMeasSwitch	NCellSrsTimeMeasSwitch	Select this option.

Table 6-10 Parameters in the **CellPdcchAlgo** MO

Parameter Name	Parameter ID	Setting Notes
The Strategy Of PDCCH Aggregation Level Adaptation	CellPdcchAlgo.PDCCHAggLvAdaptStrage	It is recommended that this parameter be set to STRATEGYBASEDONCAPACITY when D-MIMO is enabled.

Table 6-11 Parameters in the **CellCqiAdjAlgo** MO

Parameter Name	Parameter ID	Setting Notes
Cell CQI Adjustment Schedule Count Thld	CellCqiAdjAlgo.CellCqiAdjSchCntThld	50

Parameters Used for Optimization

The SRS resource optimization function is supported in LampSite inter-cell D-MIMO scenarios to improve D-MIMO performance. This function increases the proportion of JT UEs allocated with short-period SRS resources in the inter-cell D-MIMO cluster, improving CEU performance and average cell performance in the downlink.

The **SRSCfg.SrsResOptSwitch** parameter controls this function and can be configured only when all of the following conditions are met:

- Inter-cell D-MIMO is enabled.
- UBBP boards are manually bound.
- The **SRSCfg.TddSrsCfgMode** parameter is set to **ACCESS_ENHANCED**.
- The **Cell.TxRxMode** parameter is set to **2T2R**.

NOTICE

- Modifying the **SRSCfg.SrsResOptSwitch** parameter is a high-risk operation. Therefore, perform this operation with caution.
- This function requires the UBBPe/UBBPd9.
- The UBBPd4 and LBBPd boards do not support SRS resource optimization. If this function is enabled on these boards, physical cells on these boards cannot be activated, and the eNodeB reports ALM-29243 Cell Capability Degraded or ALM-29240 Cell Unavailable. In this situation, the value of the **Work Status** parameter is **The BBP configuration does not support SRS resource optimization** in the **DSP CELL** command output.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following illustrates how to enable inter-cell D-MIMO for three macro cells, with the **Cell.LocalCellId** parameter set to **0, 1, and 2**.

```
//Specifying baseband equipment for the BBP and co-processing board
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=3;
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=4;
//Adding cells and configuring PCIs
ADD CELL: LocalCellId=0, CellName="huawei", FreqBand=38, UEarfcnCfgInd=NOT_CFG, DEarfcn=37850,
UIBandWidth=CELL_BW_N100, DIBandWidth=CELL_BW_N100, CellId=1, PhyCellId=3, FddTddInd=CELL_TDD,
SubframeAssignment=SA2, SpecialSubframePatterns=SSP7, EuCellStandbyMode=ACTIVE,
RootSequenceIdx=10, MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=8T8R;
ADD CELL: LocalCellId=1, CellName="huawei", FreqBand=38, UEarfcnCfgInd=NOT_CFG, DEarfcn=37850,
UIBandWidth=CELL_BW_N100, DIBandWidth=CELL_BW_N100, CellId=2, PhyCellId=6, FddTddInd=CELL_TDD,
SubframeAssignment=SA2, SpecialSubframePatterns=SSP7, EuCellStandbyMode=ACTIVE,
RootSequenceIdx=10, MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=8T8R;
ADD CELL: LocalCellId=2, CellName="huawei", FreqBand=38, UEarfcnCfgInd=NOT_CFG, DEarfcn=37850,
UIBandWidth=CELL_BW_N100, DIBandWidth=CELL_BW_N100, CellId=3, PhyCellId=9, FddTddInd=CELL_TDD,
SubframeAssignment=SA2, SpecialSubframePatterns=SSP7, EuCellStandbyMode=ACTIVE,
RootSequenceIdx=10, MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=8T8R;
//Adding operator information for the cells
ADD CELLOP: LocalCellId=0, TrackingAreaId=0;
ADD CELLOP: LocalCellId=1, TrackingAreaId=0;
ADD CELLOP: LocalCellId=2, TrackingAreaId=0;
//Binding the cells with the Layer 1 baseband equipment
ADD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=0, BaseBandEqmId=1;
ADD EUCELLSECTOREQM: LocalCellId=1, SectorEqmId=1, BaseBandEqmId=1;
ADD EUCELLSECTOREQM: LocalCellId=2, SectorEqmId=2, BaseBandEqmId=1;
//Binding the cells with the primary baseband equipment
ADD EUCELLPRIBBEQM: LocalCellId=0, PriBaseBandEqmId=1;
ADD EUCELLPRIBBEQM: LocalCellId=1, PriBaseBandEqmId=1;
ADD EUCELLPRIBBEQM: LocalCellId=2, PriBaseBandEqmId=1;
//(Required only when D-MIMO is enabled) Adding D-MIMO co-processing resources
ADD COPROCRES: CoProcResId=0, BaseBandEqmId=0, BundlingClusterType=DMIMO,
WorkMode=COORDINATING_PROCESSING-1&BASEBAND_PROCESSING-0;
//Adding a D-MIMO cluster
ADD DMIMOCLUSTER: DMIMOClusterId=0, CoProcResId=0;
//Adding cells to the D-MIMO cluster
ADD DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=1, eNodeBId=0, Mcc="460", Mnc="01";
ADD DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=2, eNodeBId=0, Mcc="460", Mnc="01";
ADD DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=3, eNodeBId=0, Mcc="460", Mnc="01";
//(Optional) Enabling target SRS SINR optimization for beamforming UEs
MOD CELLPCALGO: LocalCellId=0, DMSrPcSinrOffset=5;
MOD CELLPCALGO: LocalCellId=1, DMSrPcSinrOffset=5;
MOD CELLPCALGO: LocalCellId=2, DMSrPcSinrOffset=5;
//Configuring the SRS automatic neighboring cell measurement switch, SRS or DMRS automatic
neighboring cell measurement event A3 offset, and neighboring cell SRS time-domain switch
MOD NCELLSRSMEASPARA: LocalCellId=0, SrsAutoNCellMeasSwitch=ON, NCellSrsMeasA3Offset=-12,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-1;
MOD NCELLSRSMEASPARA: LocalCellId=1, SrsAutoNCellMeasSwitch=ON, NCellSrsMeasA3Offset=-12,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-1;
MOD NCELLSRSMEASPARA: LocalCellId=2, SrsAutoNCellMeasSwitch=ON, NCellSrsMeasA3Offset=-12,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-1;
//Turning on the beamforming, MU beamforming, inter-cell D-MIMO, quick channel calibration, and
downlink scheduling extension switches
MOD CELLALGOSWITCH: LocalCellId=0, BfAlgoSwitch=BfSwitch-1, MuBfAlgoSwitch=MuBfSwitch-1,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-1, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-1,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-1;
MOD CELLALGOSWITCH: LocalCellId=1, BfAlgoSwitch=BfSwitch-1, MuBfAlgoSwitch=MuBfSwitch-1,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-1, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-1,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-1;
MOD CELLALGOSWITCH: LocalCellId=2, BfAlgoSwitch=BfSwitch-1, MuBfAlgoSwitch=MuBfSwitch-1,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-1, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-1,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-1;
//(Optional) Turning on the CRS conflict-related interference avoidance optimization switch when PCI mod
3 values are aligned
MOD CELLALGOSWITCH: LocalCellId=0, AvoidInterfSwitch=AvoidCrsConflictInterfSw-1;
MOD CELLALGOSWITCH: LocalCellId=1, AvoidInterfSwitch=AvoidCrsConflictInterfSw-1;
```

```

MOD CELLALGOSWITCH: LocalCellId=2, AvoidInterfSwitch=AvoidCrsConflictInterfSw-1;
//Configuring DmimoA3RsrpOffset and DmimoJtCellRsrpDiffThld
MOD DMIMOALGO: LocalCellId=0, DmimoA3RsrpOffset=-12, DmimoJtCellRsrpDiffThld=-20;
MOD DMIMOALGO: LocalCellId=1, DmimoA3RsrpOffset=-12, DmimoJtCellRsrpDiffThld=-20;
MOD DMIMOALGO: LocalCellId=2, DmimoA3RsrpOffset=-12, DmimoJtCellRsrpDiffThld=-20;
//Configuring the reserved resource SRS periods 1 and 2
MOD SRSCFG: LocalCellId=0, RsvResSrsPeriod1=10ms, RsvResSrsPeriod2=20ms;
MOD SRSCFG: LocalCellId=1, RsvResSrsPeriod1=10ms, RsvResSrsPeriod2=20ms;
MOD SRSCFG: LocalCellId=2, RsvResSrsPeriod1=10ms, RsvResSrsPeriod2=20ms;
//Configuring CEU-related counter parameters
MOD CELLCOUNTERPARAGROUP: LocalCellId=0, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-1,
EdgeUserA3Offset=-12;
MOD CELLCOUNTERPARAGROUP: LocalCellId=1, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-1,
EdgeUserA3Offset=-12;
MOD CELLCOUNTERPARAGROUP: LocalCellId=2, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-1,
EdgeUserA3Offset=-12;
//Adding neighbor relationships
ADD EUTRANINTRAFFREQNCELL: LocalCellId=0, Mcc="460", Mnc="01", eNodeBId=0, CellId=2;
ADD EUTRANINTRAFFREQNCELL: LocalCellId=1, Mcc="460", Mnc="01", eNodeBId=0, CellId=1;
ADD EUTRANINTRAFFREQNCELL: LocalCellId=0, Mcc="460", Mnc="01", eNodeBId=0, CellId=3;
ADD EUTRANINTRAFFREQNCELL: LocalCellId=2, Mcc="460", Mnc="01", eNodeBId=0, CellId=1;
ADD EUTRANINTRAFFREQNCELL: LocalCellId=1, Mcc="460", Mnc="01", eNodeBId=0, CellId=3;
ADD EUTRANINTRAFFREQNCELL: LocalCellId=2, Mcc="460", Mnc="01", eNodeBId=0, CellId=2;
//Optional) Configuring the PDCCH aggregation level adaptation policy
MOD CELLPDCCHALGO: LocalCellId=0, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCAPACITY;
MOD CELLPDCCHALGO: LocalCellId=1, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCAPACITY;
MOD CELLPDCCHALGO: LocalCellId=2, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCAPACITY;
//Changing the cell-level CQI adjustment scheduling count threshold
MOD CELLCQIADJALGO: LocalCellId=0, CellCqiAdjSchCntThld = 50;
MOD CELLCQIADJALGO: LocalCellId=1, CellCqiAdjSchCntThld = 50;
MOD CELLCQIADJALGO: LocalCellId=2, CellCqiAdjSchCntThld = 50;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

//Optional) Disabling target SRS SINR optimization for beamforming UEs
MOD CELLPCALGO: LocalCellId=0, DMSrsPcSinrOffset=0;
MOD CELLPCALGO: LocalCellId=1, DMSrsPcSinrOffset=0;
MOD CELLPCALGO: LocalCellId=2, DMSrsPcSinrOffset=0;
//Turning off the SRS auto neighbor cell measurement switch and the neighboring cell SRS time-domain
measurement switch and restoring the SRS or DMRS auto neighbor cell measurement A3 offset to the
value used before inter-cell D-MIMO is enabled
MOD NCELLSRSMESAPARA: LocalCellId=0, SrsAutoNCellMeasSwitch=OFF, NCellSrsMeasA3Offset=-20,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-0;
MOD NCELLSRSMESAPARA: LocalCellId=1, SrsAutoNCellMeasSwitch=OFF, NCellSrsMeasA3Offset=-20,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-0;
MOD NCELLSRSMESAPARA: LocalCellId=2, SrsAutoNCellMeasSwitch=OFF, NCellSrsMeasA3Offset=-20,
NCellMeasSwitch=NCellSrsTimeMeasSwitch-0;
//Turning off the beamforming, MU beamforming, inter-cell D-MIMO, quick channel calibration, and
downlink scheduling extension switches
MOD CELLALGOSWITCH: LocalCellId=0, BfAlgoSwitch=BfSwitch-0, MuBfAlgoSwitch=MuBfSwitch-0,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-0, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-0,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-0;
MOD CELLALGOSWITCH: LocalCellId=1, BfAlgoSwitch=BfSwitch-0, MuBfAlgoSwitch=MuBfSwitch-0,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-0, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-0,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-0;
MOD CELLALGOSWITCH: LocalCellId=2, BfAlgoSwitch=BfSwitch-0, MuBfAlgoSwitch=MuBfSwitch-0,
DMIMOAlgoSwitch=InterCellDmimoJTSwitch-0, EnhChnCalSwitch=QUICK_CHN_CAL_SWITCH-0,
DISchExtSwitch=DL_COORD_SCH_EXP_OPT_SW-0;
//Optional) Turning off the CRS conflict-related interference avoidance optimization switch when PCI mod
3 values are aligned

```



```
MOD CELLALGOSWITCH: LocalCellId=0, AvoidInterfSwitch=AvoidCrsConflictInterfSw-0;
MOD CELLALGOSWITCH: LocalCellId=1, AvoidInterfSwitch=AvoidCrsConflictInterfSw-0;
MOD CELLALGOSWITCH: LocalCellId=2, AvoidInterfSwitch=AvoidCrsConflictInterfSw-0;
//Removing a cell from the D-MIMO cluster
RMV DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=0, eNodeBId=0, Mcc="100", Mnc="01";
RMV DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=1, eNodeBId=0, Mcc="100", Mnc="01";
RMV DMIMOCLUSTERCELL: DMIMOClusterId=0, CellId=2, eNodeBId=0, Mcc="100", Mnc="01";
//Removing the D-MIMO cluster
RMV DMIMOCLUSTER: DMIMOClusterId=0;
//Restoring reserved resource SRS period 2 to the value used before inter-cell D-MIMO is enabled
MOD SRSCFG: LocalCellId=0, RsvResSrsPeriod2=10ms;
MOD SRSCFG: LocalCellId=1, RsvResSrsPeriod2=10ms;
MOD SRSCFG: LocalCellId=2, RsvResSrsPeriod2=10ms;
//Restoring CEU-related counter parameters
MOD CELLCOUNTERPARAGROUP: LocalCellId=0, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-0,
EdgeUserA3Offset=-13;
MOD CELLCOUNTERPARAGROUP: LocalCellId=1, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-0,
EdgeUserA3Offset=-13;
MOD CELLCOUNTERPARAGROUP: LocalCellId=2, CellCounterAlgoSwitch=BasedA3EdgeUserSwitch-0,
EdgeUserA3Offset=-13;
//((Optional) Restoring the PDCCH aggregation level adaptation policy
MOD CELLPDCCHALGO: LocalCellId=0, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCOVERAGE;
MOD CELLPDCCHALGO: LocalCellId=1, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCOVERAGE;
MOD CELLPDCCHALGO: LocalCellId=2, PDCCHAggLvAdaptStrage=STRATEGYBASEDONCOVERAGE;
//Restoring the cell-level CQI adjustment scheduling count threshold to the value used before D-MIMO is
enabled
MOD CELLCQIADJALGO: LocalCellId=0, CellCqiAdjSchCntThld = 0;
MOD CELLCQIADJALGO: LocalCellId=1, CellCqiAdjSchCntThld = 0;
MOD CELLCQIADJALGO: LocalCellId=2, CellCqiAdjSchCntThld = 0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

- Step 1** Run the **DSP CELL** and **LST CELL** commands to check the cell status of the D-MIMO cluster. The cells in the D-MIMO cluster are working properly if all of the following conditions are met in the command outputs:
 - The value of the **Cell instance state** parameter is **Normal**, indicating that the cells have been activated.
 - The value of the **Work Status** parameter is **Normal** for all the RRUs, indicating that the RRUs are working properly.
- Step 2** Run the **DSP CELLCALIBRATION** command to check the channel calibration status. Channel calibration is successful if the value of the **Calibration Result** is **Success**.
- Step 3** Run the **DSP DMIMOCLUSTERCELL** command to check the cell status of the D-MIMO cluster. The D-MIMO cluster is working properly if the value of the **DMIMO Cluster Cell Status** parameter is **Normal**.
- Step 4** If the cells in a D-MIMO cluster are working properly, run the **DSP DMIMOCALIBRATION** command to display the channel calibration status of the D-MIMO cluster. If the value of the **D-MIMO Calibration Result** parameter is **Succeeded**, channel calibration is successful in the D-MIMO cluster. In this

situation, SU-JT and MU-JT are available. If the value is **Failed**, channel calibration in the D-MIMO cluster failed. In this case, non-coherent joint transmission can be performed, but not SU-JT or MU-JT.

 NOTE

The D-MIMO cluster is working properly if the value of the **D-MIMO Cluster Cell Status** parameter is **Normal**. The D-MIMO cluster works properly even when channel calibration fails in the D-MIMO cluster. In this situation, only non-coherent joint transmission can be performed, but not SU-JT or MU-JT.

----End

6.4.3 Network Monitoring

Monitoring Whether D-MIMO Takes Effect

The following counters can be used on the eNodeB side to check whether D-MIMO has taken effect.

Table 6-12 Counters used to observe UE pairing for D-MIMO

Counter Name	Counter Description
L.ChMeas.DMIMO.2Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 2
L.ChMeas.DMIMO.3Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 3
L.ChMeas.DMIMO.4Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 4
L.ChMeas.DMIMO.5Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 5
L.ChMeas.DMIMO.6Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 6
L.ChMeas.DMIMO.7Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 7
L.ChMeas.DMIMO.8Layer.PairPRB	Average number of PRBs paired for D-MIMO at layer 8
Note: The counters measure the number of PRBs paired for D-MIMO at the corresponding layers.	

Table 6-13 Counters used to observe the proportion and scope of UEs on which D-MIMO takes effect

Counter Name	Counter Description
L.ChMeas.DMIMO.JT.User.Avg	Number of D-MIMO JT UEs in the downlink
L.ChMeas.DMIMO.JTUser.RRU.Avg	Average number of working RRUs for D-MIMO JT UEs in the downlink
Note: Inter-cell D-MIMO has taken effect if the value of the L.ChMeas.DMIMO.JT.User.Avg counter is greater than 0.	

Monitoring the Working Status and Channel Calibration Status of the D-MIMO Cluster

The external CHR counter **PERIOD_CLUSTER_DMIMO_MR** is used to monitor the working status and the channel calibration status of the D-MIMO cluster.

This counter periodically measures the invalidity duration of the D-MIMO cluster. The working status of the D-MIMO cluster can be determined based on the invalidity duration. The default measurement period is 15 minutes.

Field	Description
>DmimoClusterID	ID of the D-MIMO cluster to which the logical cells performing MR reporting belong
>ClusterAbnormallatency	Invalidity duration of the D-MIMO cluster
>CalibrationAbnormallatency	Channel calibration invalidity duration of the D-MIMO cluster

Monitoring the Pairing Status in a D-MIMO Cell

On the MAE-Access, the following **MIMO (Cell) Monitoring** items can be used to monitor the pairing status in a D-MIMO cell.

Monitoring Item	Unit	Description
Number of successful DMIMO Pairing TM7 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM7 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$

Monitoring Item	Unit	Description
Number of successful DMIMO Pairing TM8 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM8 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$
Number of successful DMIMO Pairing TM9 RB(Num) with N Layers	Number	Number of RBs that are successfully paired for D-MIMO in TM9 at layer N within a monitoring period. The total number of available RBs that are successfully paired for D-MIMO during all TTIs is averaged at the end of the monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$
JT User Num	Number	Average number of JT UEs in a cell in each TTI within a monitoring period
JT User Work RRU Num	Number	Average number of working RRUs for JT UEs in the local cell within a monitoring period
SUJT Schedule RB Num	Number	Accumulative number of PRBs scheduled for JT UEs in SU-JT mode in a cell within a monitoring period

Monitoring the Pairing Status of a Single UE in a D-MIMO Cell

On the MAE-Access, the following **MIMO(User) Monitoring** items can be used to monitor the pairing status of a single UE in a D-MIMO cell.

Monitoring Item	Unit	Description
Number of successful DMIMO pairing RB (Num) with N Layers	Number	Number of RBs for UEs successfully paired for D-MIMO at layer N within a monitoring period. The total number of RBs for UEs successfully paired for D-MIMO at layer N during all TTIs is averaged as a cell-level value (including all RRUs) at the end of a monitoring period. $N = \{2, 3, 4, 5, 6, 7, 8\}$ This item is measured only when the Test Items parameter is set to TDD DMIMO . Value range: 0–1000
DMIMO User Indication	None	User attributes, for example, non-JT or JT UEs 0: non-JT UE 1: JT UE 255: invalid value
JT User Work RRU Num	Number	Average number of working RRUs for JT UEs within a monitoring period
SUJT Schedule RB Num	Number	Number of RBs scheduled for JT UEs in SU-JT mode within a monitoring period

Monitoring CEU Performance

The following counters can be used to monitor CEU performance on the eNodeB side.

Counter Name	Counter Description
L.Thrp.bits.UL.BorderUE	Uplink PDCP data volume for CEUs in a cell
L.Thrp.bits.UL.SmallPkt.BorderUE	Uplink PDCP data volume for CEUs' small packets in a cell
L.Thrp.Time.UL.RmvSmallPkt.BorderUE	Uplink data transmission duration for CEUs' non-small packets in a cell
L.Thrp.bits.DL.BorderUE	Downlink PDCP data volume for CEUs in a cell
L.Thrp.bits.DL.LastTTI.BorderUE	Downlink PDCP data volume for CEUs in the last TTI before the buffer is empty in a cell

Counter Name	Counter Description
L.Thrp.Time.DL.RmvLastTTI.BorderUE	Downlink data transmission duration for CEUs except the last TTI before the buffer is empty in a cell

6.4.4 Possible Issues

A D-MIMO cell in a D-MIMO cluster will automatically revert to a normal cell when a fault related to activation of a cell in a D-MIMO cluster occurs, such as RRU faults, including RRU antenna rollback. The D-MIMO cell does not revert when a channel calibration fault occurs. If this happens, only non-coherent joint transmission, not coherent joint transmission, can be performed for UEs in the D-MIMO cluster.

Channel Calibration Failures

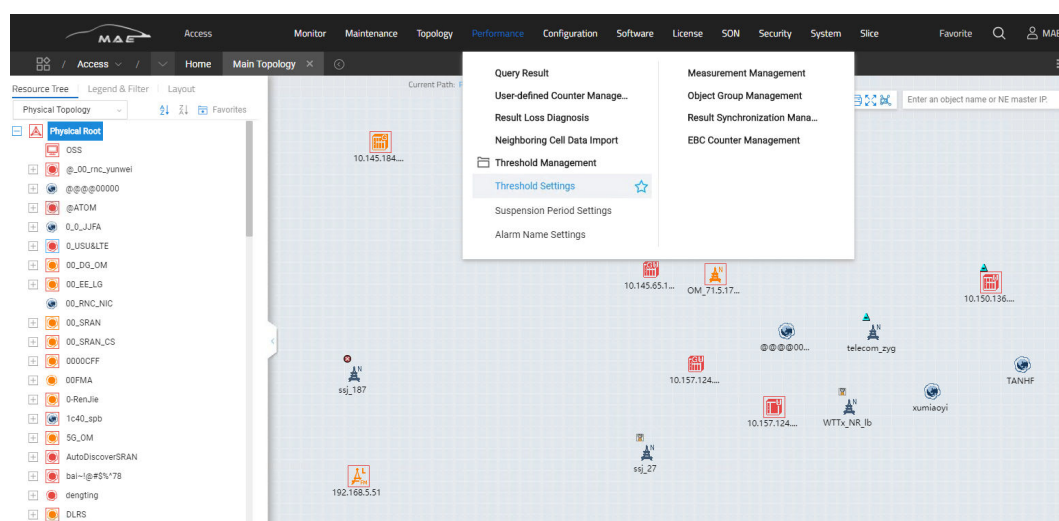
A user-defined performance threshold alarm can be configured on the MAE-Access. The alarm is reported when the duration of the channel calibration failure exceeds the configured threshold. The alarm helps O&M personnel more quickly identify channel calibration failures.

NOTE

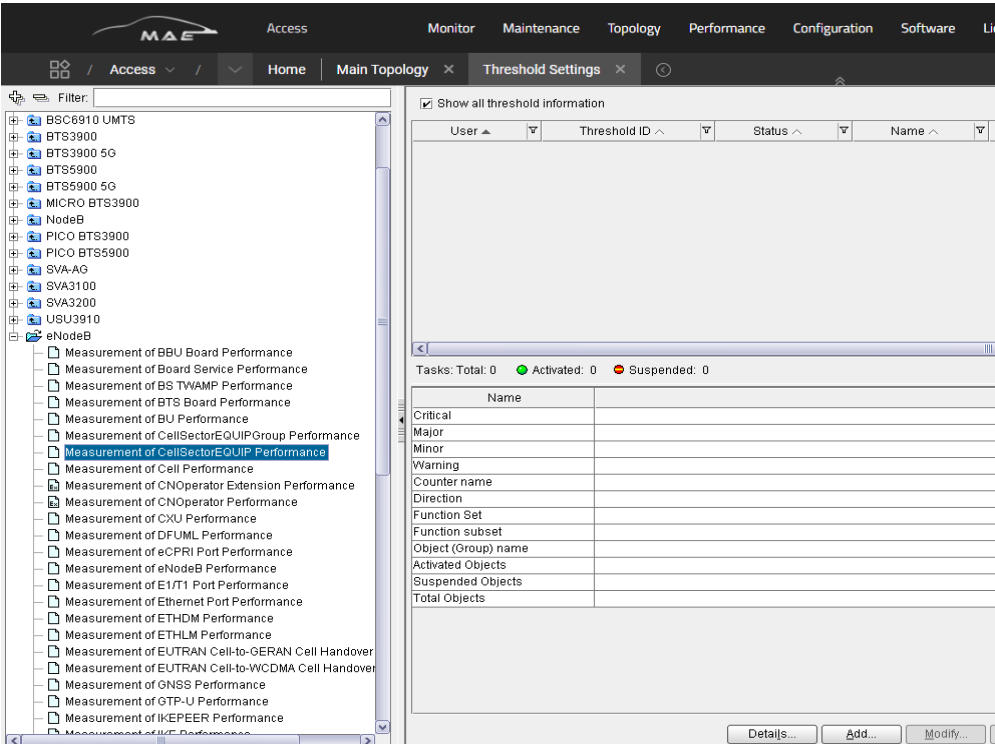
The **L.CellSectorEqpt.UNA.Dur.Cali** counter measures the channel calibration failure duration.

To define a channel calibration failure alarm, perform the following operations on the MAE-Access:

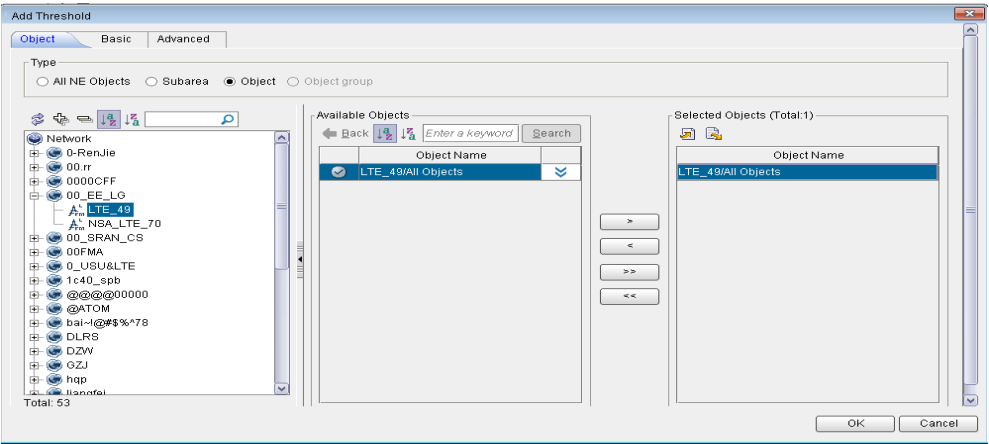
- Step 1** On the MAE-Access, choose **Performance > Threshold Management > Threshold Settings**. The **Threshold Settings** tab page is displayed.



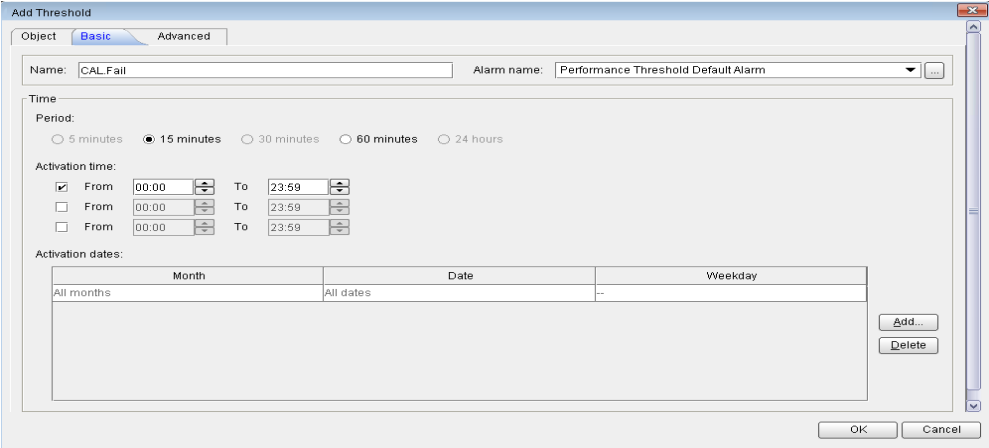
- Step 2** Choose **eNodeB > Measurement of CellSectorEQUIP Performance** from the navigation tree, and then click **Add** at the bottom right part of the tab page. The **Add Threshold** dialog box is displayed.



Step 3 Select some or all NEs in the **Object** tab of the **Add Threshold** dialog box.



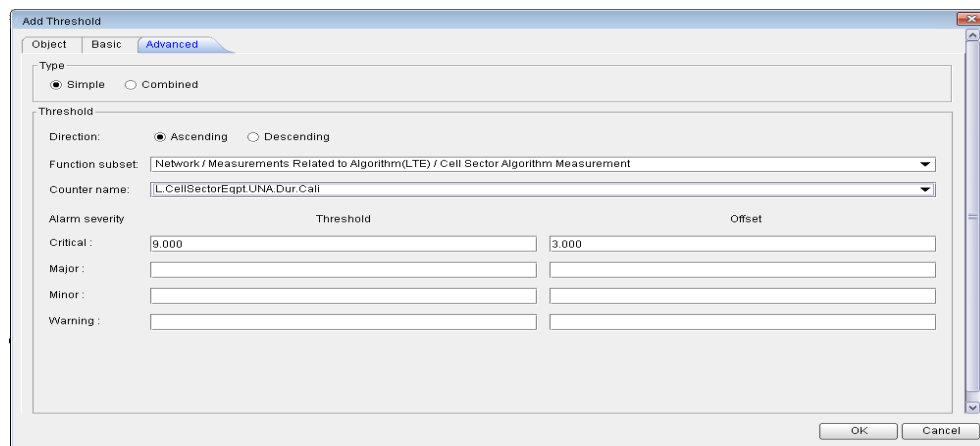
Step 4 On the **Basic** tab page, set **Name**, **Period**, and **Activation time**.



Step 5 On the **Advanced** tab page, perform the following operations and then click **OK**.

An alarm is reported if the duration of the channel calibration failure is greater than or equal to the threshold plus the offset. This alarm is cleared if the duration falls below the threshold minus the offset.

- Set **Direction** to **Ascending**.
- Set **Function subset** to **Network / Measurements Related to Algorithm(LTE) / Cell Sector Algorithm Measurement**.
- Set **Counter name** to **L.CellSectorEqpt.UNA.Dur.Cali**.
- Set **Threshold** based on the calculation result of **Period** set on the **Basic** tab page multiplied by 80% minus the offset. In the example, set **Threshold** to **9**, namely $15 \times 80\% - 3$.
- Set **Offset** based on the calculation result of **Period** set on the **Basic** tab page multiplied by 20%. In the example, set **Offset** to **3**, namely $15 \times 20\%$.



----End

Channel Calibration-related Troubleshooting

If no user-defined performance threshold alarms are reported on the MAE-Access, run the **DSP DMIMOCALIBRATION** command to query the channel calibration status. In the command output, if the value of the **D-MIMO Calibration Result** parameter is not **Succeeded**, there are the following other possible values and actions to be taken:

- If the value is **Exception occurred in internal calibration**, intra-RRU channel calibration has failed. Run the **DSP CELLCALIBRATION** command to identify the faulty RRU and then contact Huawei technical support for fault handling.
- If the value is **Route search failed**, no appropriate routing path has been found between RRUs because the RRU experiences external interference. The interference needs to be eliminated first.
- If the value is **Reciprocity calibration failed**, the intra-cluster RRU route search was successful but inter-RRU reciprocity calibration failed due to the failure to reach specified thresholds. Obtain the values of the **Calibration Signal CINR** and **Calibration Signal RSSI** parameters and then contact Huawei technical support for troubleshooting.

Faults Related to Activation of Cells in a D-MIMO Cluster

The value of the **D-MIMO Cluster Cell Status** parameter is not **Normal** in the **DSP DMIMOCCLUSTERCELL** command output.

- If the value is **incorrect configurations** or **unavailable licenses**, check whether the following conditions are met: 1. The inter-cell D-MIMO switch is turned on. 2. The number of RRUs is configured properly. 3. The inter-cell D-MIMO cluster is configured properly. 4. The license has been activated. 5. Cells in the inter-cell D-MIMO cluster have consistent parameter settings, including the number of CRS ports, TDD SRS configuration mode, reserved SRS periods 1 and 2, and subframe configuration.
- If the value is **Limited hardware capacity**, check whether the cells in the inter-cell D-MIMO cluster are set up on the same UBBPd9 or UBBPe.
- If the value is **Route application failures**, check whether the routing bandwidth is sufficient.
- If the value is **Cell abnormal**, run the **DSP CELL** command to verify the cell status.
- If the value is **Insufficient co-processing resources**, check whether LCOP configurations meet the requirements provided in [6.3.3 Hardware](#).
- If the value is **Channel calibration failures**, run the **DSP DMIMOCALIBRATION** command to identify the cause.
- If the value is **Cluster ID conflicts**, check whether the cluster ID is configured according to the restraints provided in [6.4.1.1 Data Preparation](#).
- If the value is **Clusters being established**, wait until the cluster is set up successfully.
- If the value is **Co-processing resources abnormal**, check whether co-processing resources of the inter-cell D-MIMO cluster are configured on the LCOP.
- If the value is **The number of LampSite cells in a cluster must be greater than the specified minimum number to enable D-MIMO**, check whether there are two or more cells in the cluster.
- If the value is **LampSite cells cannot form a D-MIMO cluster with cells of other types of eNodeBs that have different PCIs**, check whether only LampSite cells are configured in the cluster.
- If the value is **TM8 dual-stream beamforming is configured for micro eNodeBs**, check whether the **CellBf.MaxBfRankPara** parameter is set to **DUAL_LAYER_BF** for cells set up on micro eNodeBs. If yes, the cluster is abnormal.

Alarm Reported

If inter-cell D-MIMO is activated and either of the alarms listed in [Table 6-14](#) is reported, handle the alarm according to the alarm reference. In addition, check whether the BBP corresponding to the baseband equipment that provides co-processing resources supports centralized control for inter-cell D-MIMO. If not, replace the BBP. For details, see [6.3.3 Hardware](#).

Table 6-14 eNodeB alarms related to BBPs

Alarm ID	Alarm Name
26245	Configuration Data Inconsistency
26203	Board Software Program Error

Others

When the **DmimoJTSwitch** option of the **CellAlgoSwitch.DMIMOAlgoSwitch** parameter is selected, consecutive block errors may arise during dual-stream beamforming, even leading to network disconnections. For details about the workarounds, see *Beamforming (TDD)*.

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter.

View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

1. *SFN*
2. *Energy Conservation and Emission Reduction*
3. *Beamforming (TDD)*
4. *DRX and Signaling Control*
5. *Extended CP*
6. *High Speed Mobility*
7. *Relay*
8. *Massive MIMO Basics (TDD)*
9. *Massive MIMO Enhancements (TDD)*
10. *TMA (TDD)*
11. *WTTx Turbo Beamforming (TDD)*
12. *DL CoMP (TDD)*
13. *Downlink Scheduling*
14. *Uplink Coordinated Scheduling*
15. *RAN Sharing*
16. *Soft Split Resource Duplex (TDD)*
17. *Dynamic Power Sharing Between LTE Carriers*
18. *MIMO*
19. *LampSite BBU Technical Specifications*
20. *Cloud BB Overview*
21. *USU3910-based Multi-BBU Interconnection*
22. *Massive MIMO Optimization in WTTx Scenarios (TDD)*
23. *Intelligent Multi-Beam of 8T8R (TDD)*
24. *Carrier Aggregation*
25. *Massive MIMO Optimization in WTTx Scenarios (TDD)*
26. *Green Site*